Software Design — Vehicle Washing Management System

Cover Page

Table 1

Document Identification Data

Field	Value
Project	Vehicle Washing Management System (Vehicle Washing / LavaRápido)
Deliverable	Software design documentation (section 2.5.4)
Program	Software Analysis and Development Technologist (ADSO)
Group (ficha)	3145556
Apprentices	Brayan Estiven Patiño Cabrera (project lead), Juan Camilo Guilombo Osorio (frontend lead), Merari Urbano (backend lead)
Instructor	Karol Daniela Correa
Training center	Industria
Date	October 27, 2026
Version	1.11

> *Note.* Cover data was taken from the project's Technical Manual v2.0. All design content was derived from the source code in the `Sistema Lavado Vehicular` repository (Backend, Web, and LavadoVehicular-Movil) and must be validated with the instructor.

1. Introduction and Scope

The system consists of seven backend services behind an API Gateway, a web application in Angular 21, and a mobile application in React Native (Expo 57); both clients serve three roles: customer, operator, and administrator.

This document describes the design as built: user interfaces, architectural model, design patterns, data model, and main classes. Every statement cites the file where it can be verified. Diagrams are written in Mermaid (Mermaid, n.d.) using UML notation. File paths are relative to the `Sistema Lavado Vehicular/` folder.

Table 2

System Components and Technology

Component	Technology	Port	Folder
API Gateway	Spring Boot 4.0.8, Spring Cloud Gateway (WebMVC), Java 21	8080	`Backend/lavarapido-api-gateway`
security-service	Spring Boot 4.0.8, Spring Security, JWT, Java 21	3001	`Backend/security-service`
customer-service	Spring Boot 4.0.8, JPA, Java 21	3002	`Backend/lavarapido-customer-service`
booking-service	Spring Boot 4.0.8, JPA, Java 21	3003	`Backend/lavarapido-booking-service`
operations-service	Spring Boot 4.0.8, JPA, Java 21	3004	`Backend/lavarapido-operations-service`
payment-service	ASP.NET Core 8, Entity Framework Core 8	3005	`Backend/lavarapido-payment-service`
notification-service	Spring Boot 4.0.8, Spring Mail, Java 21	3006	`Backend/lavarapido-notification-service`
Database	SQL Server 2022 (one database, one schema per service)	1433	`Backend/lavarapido-infra/docker-compose.yml`
Messaging	RabbitMQ 4 (exchange `carwash.events`)	5672	`Backend/lavarapido-infra/docker-compose.yml`
Web application	Angular 21.2.25, Angular Material, ngx-translate, Chart.js	4200	`Web`
Mobile application	React Native 0.86.3, Expo ~57, React Navigation 7, i18next	8081 (Expo)	`LavadoVehicular-Movil`

> *Note.* Versions taken from `pom.xml`, `Directory.Build.props`, and `package.json`. Official documentation: Spring (n.d.), Microsoft (n.d.-a, n.d.-b), RabbitMQ (n.d.), Liquibase (n.d.), Angular (n.d.), React Native (n.d.), and Expo (n.d.).

Out of scope: requirements (see SRS v7, `lavarapido-docs/01_INFORME_DE_ESPECIFICACION_DE_REQUISITOS`), dynamic diagrams (analysis deliverable), and costs.

2. User Interfaces

The web client has 35 routed screens and the mobile app has 31; both split the areas by role and only let each user into their own. On the web this is done by `authGuard` with `data.roles` (`Web/src/app/app.routes.ts:19-49`). On mobile, `RootNavigator` registers only the screens of the logged-in role (`LavadoVehicular-Movil/src/core/navigation/RootNavigator.tsx:63-110`).

2.1 Web Application Screen Inventory

Table 3

Web Application Screens (Angular)

Route	Screen	Role	Purpose	Services / endpoints consumed
`/`	Landing	Public	Presents the car wash, its services, the app download, contact info, and frequently asked questions	`GET /establishment` (business email and phone)
`/auth`	Login	Public	Authenticate with email and password	`POST /auth/login`
`/auth/register`	Sign-up	Public	Create a customer account	`POST /auth/register`
`/auth/forgot-password`	Password recovery (3 steps)	Public	Email → code → new password	`POST /auth/password/forgot`, `/verify`, `/reset`
`/report/:token`	Inspection report	Public	View the report shared by the admin	`GET /public/inspections/{token}` and its photos
**	Page not found	Public	Tell that the address does not exist and go back home	None
`/client`	Customer home	CLIENT	Summary, next booking with its operator, and points	`GET /bookings/me`, `/vehicles`, `/loyalty/balance`, `/operators/my-bookings`
`/client/reserve`	Book	CLIENT	Choose vehicle, service, date, and time	`GET /catalog/services`, `/bookings/availability`, `/operators/available`; `POST /bookings`

Route	Screen	Role	Purpose	Services / endpoints consumed
`/client/payment`	Payments	CLIENT	Pay for a booking and upload a receipt (compressed to at most 2 MB before sending; the QR can be enlarged with a magnifier); Cash only appears while its account is active; once the payment is approved only the summary is left	`GET /payment-accounts`, `/payments/me/bookings`; `POST /payments`; `/loyalty/*`
`/client/history`	History	CLIENT	Bookings from newest to oldest, payments, cancellation, and rating; clicking one opens its detail with **Go to payment**	`/bookings/me`, `POST /bookings/{id}/cancel`, `/payments/me`, `POST /ratings/{bookingId}`
`/client/vehicles`	My vehicles	CLIENT	Vehicle CRUD	`GET/POST/PUT/DELETE /vehicles`, `GET /vehicle-types`
`/client/notifications`	Notifications	CLIENT	Notification inbox; clicking one opens its detail with **Go to...**	`/notifications/*`
`/client/profile`	Profile	CLIENT	Personal data, address, email, and password	`/users/me`, `/users/me/address`, `/users/me/password`, `/users/me/email`
`/client/configuration`	Settings	CLIENT	Theme, language, and notification channels	`/users/me/preferences`

Route	Screen	Role	Purpose	Services / endpoints consumed
`/admin`	Dashboard	ADMIN	Daily indicators and alerts, bookings without an operator, and the "To close" card (only when something is pending)	`GET /admin/bookings/day-summary`, `GET /admin/assignments/unassigned`, `GET /admin/assignments/to-close`, `POST /admin/assignments/{bookingId}/attended`, and booking, payment, operator, and schedule stores
`/admin/reservations`	Bookings	ADMIN	Create, reschedule, change status, and assign an operator	`/admin/bookings/*`, `/admin/assignments/*`
`/admin/reservations/:id/inspection`	Inspection	ADMIN	Go through the phases, record findings with photos, publish and share	`/admin/inspections/{bookingId}/**`
`/admin/payments`	Payments	ADMIN	Review, approve, reject, and refund	`/admin/payments/*`
`/admin/management`	Management	ADMIN	Users (create, activate or deactivate; Enabled and Disabled filters), fixed roles and permission matrix (for reference and read-only since October 24, 2026: the role decides the access), services, promotions	`/admin/users`, `/admin/roles-permissions`, `/admin/catalog/services`, `/admin/promotions`
`/admin/schedule`	Schedule and bays	ADMIN	Weekly hours, exceptions, and bays	`/admin/schedule/*`, `/admin/bays/*`
`/admin/operators`	Operators	ADMIN	Operator list, team summary, and creation	`/admin/operators`, `/admin/operators/summary`, `POST /admin/users`

Route	Screen	Role	Purpose	Services / endpoints consumed
`/admin/operators/:id`	Operator detail	ADMIN	Availability, absences, profile, and weekly metrics	`/admin/operators/{id}/availability`, `/absences`, `/metrics`, `/admin/operators/profiles`, `PUT /admin/operators/{id}/profile`
`/admin/operators/:id/calendar`	Operator calendar	ADMIN	Operator agenda	`/admin/assignments`
`/admin/reports`	Reports	ADMIN	Revenue and services done (only completed bookings count); export PDF/Excel	Payment and booking stores (jsPDF, xlsx)
`/admin/notifications`	Notifications	ADMIN	Notification inbox; the detail goes to Bookings or home	`/notifications/*`
`/admin/profile`	Profile	ADMIN	Account data	`/users/me`
`/admin/settings`	Settings	ADMIN	Business data and payment methods	`/admin/payment-accounts`, `GET/PUT /admin/establishment`
`/operator`	Operator home	OPERATOR	Pending services and statistics	`/operator/services`, `/notifications`
`/operator/assigned-services`	Assigned services	OPERATOR	Start and finish services	`POST /operator/services/{id}/start`, `/finish`
`/operator/schedule`	Agenda	OPERATOR	Service calendar	`GET /operator/services`
`/operator/service-history`	History	OPERATOR	Completed services, newest first; the detail goes to Ratings if the service was rated	`GET /operator/services`
`/operator/qualifications`	Ratings	OPERATOR	Scores and comments received	`GET /operator/ratings`
`/operator/notifications`, `/profile`, `/settings`	Shared	OPERATOR	Notices, profile, and preferences	`/notifications/*`, `/users/me/*`

Note. Every API route has the `/api/v1` prefix and goes through the gateway (`Web/proxy.conf.json`). Source: `Web/src/app/core/services/*.ts` and routes in `features/*/*-routing-module.ts`.

2.2 Mobile Application Screen Inventory

The mobile app mirrors the web areas with its own React Native screens. It has no landing page or public report; the administrator's inspection report is available since October 17, 2026 (`AdminInspection`, opened from the booking detail).

Table 4

Mobile Application Screens (React Native / Expo)

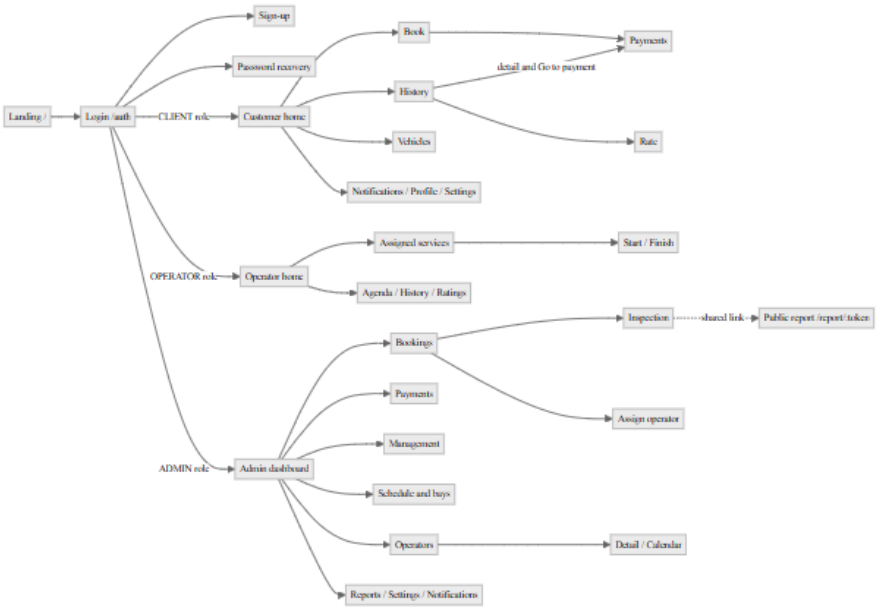
Role	Screens (`src/modules/<role>/screens`)	Web equivalent
Logged out	`LoginScreen`, `RegisterScreen`, `ForgotPasswordScreen`	`/auth/*`
CLIENT	`ClientHome`, `ClientReserve`, `ClientPayment`, `ClientHistory`, `ClientVehicles`, `ClientNotifications`, `ClientProfile`, `ClientSettings`	`/client/*`
OPERATOR	`OperatorHome`, `OperatorAssigned`, `OperatorSchedule`, `OperatorHistory`, `OperatorRatings`, `OperatorNotifications`, `OperatorProfile`, `OperatorSettings`	`/operator/*`
ADMIN	`AdminDashboard`, `AdminReservations`, `AdminPayments`, `AdminManagement`, `AdminSchedule`, `AdminOperators`, `AdminOperatorCalendar`, `AdminReports`, `AdminNotifications`, `AdminProfile`, `AdminSettings`, `AdminInspection`	`/admin/*` (`AdminInspection` = `/admin/reservations/:id/inspection`)

> *Note.* If a user has several roles, the mobile app opens the most privileged one: ADMIN, then OPERATOR, then CLIENT (`RootNavigator.tsx:42-46`). The session is stored in the system keychain with `expo-secure-store` (`src/core/storage/sessionStorage.ts:15-20`). Since October 21, 2026, when the installed APK opens, `AppUpdatesCheck` reads `/app/version.json` from the server; if it has a higher `versionCode` than the installed one, it shows a screen on top of everything that cannot be closed and only lets the user download the new APK. If the server does not answer, the app goes on as usual (`src/app/providers/AppUpdatesCheck.tsx`). It is not a routed screen, so the count of 31 does not change.

2.3 Navigation Diagram

Figure 1

Web Application Navigation by Role



The diagram shows web navigation; the mobile app follows the same path per role, without the landing page or the public report (the inspection opens from the booking detail).

2.4 Screenshots

Screenshots were taken again on October 19, 2026, with the full system running: the web client in Chrome at 1366 × 768 and the mobile app in its web build (Expo) at phone size (390 × 844). The data shown are demo accounts and bookings. Figure 12 was taken again on October 20, after the user type change and the Registered filter were removed.

Figure 2

Public Landing Page (Web)

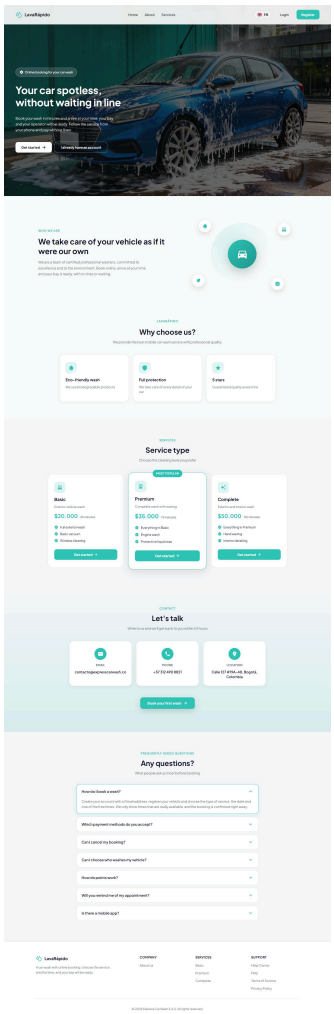


Figure 3

Login (Web)

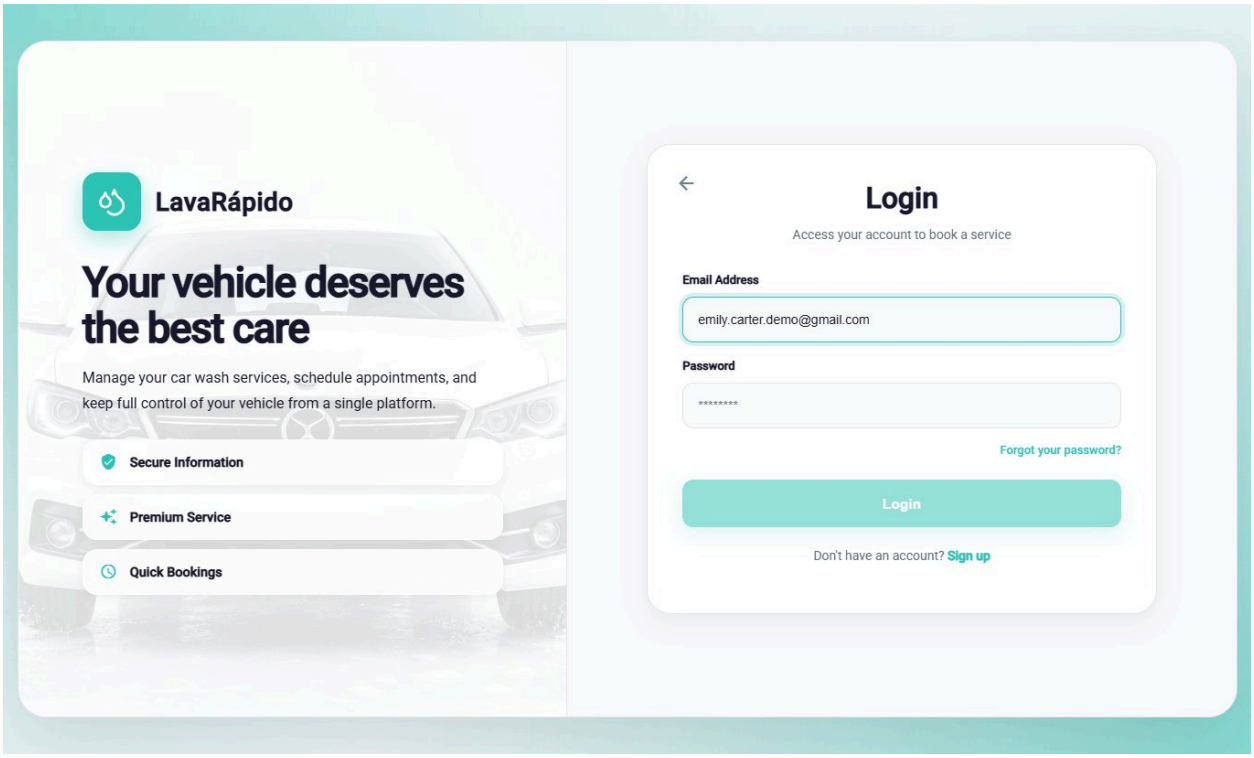


Figure 4

Login (Mobile App)

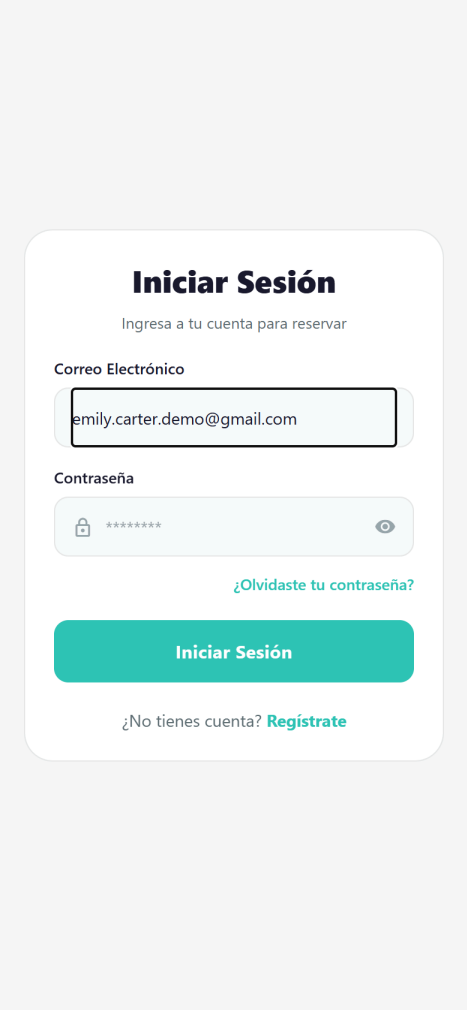


Figure 5

Book a Wash — Customer (Web)

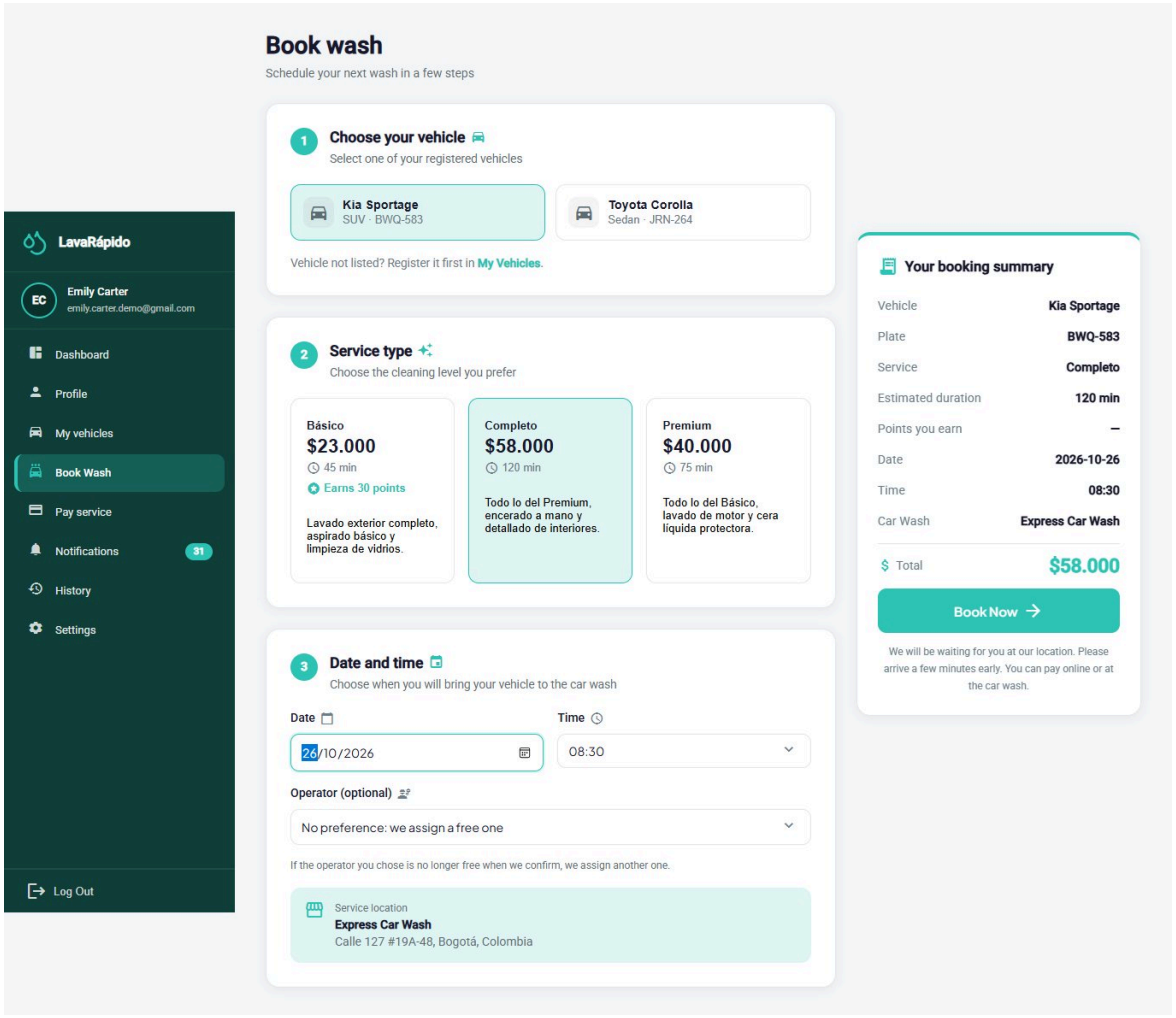


Figure 6

Payments With QR and Receipt — Customer (Web)

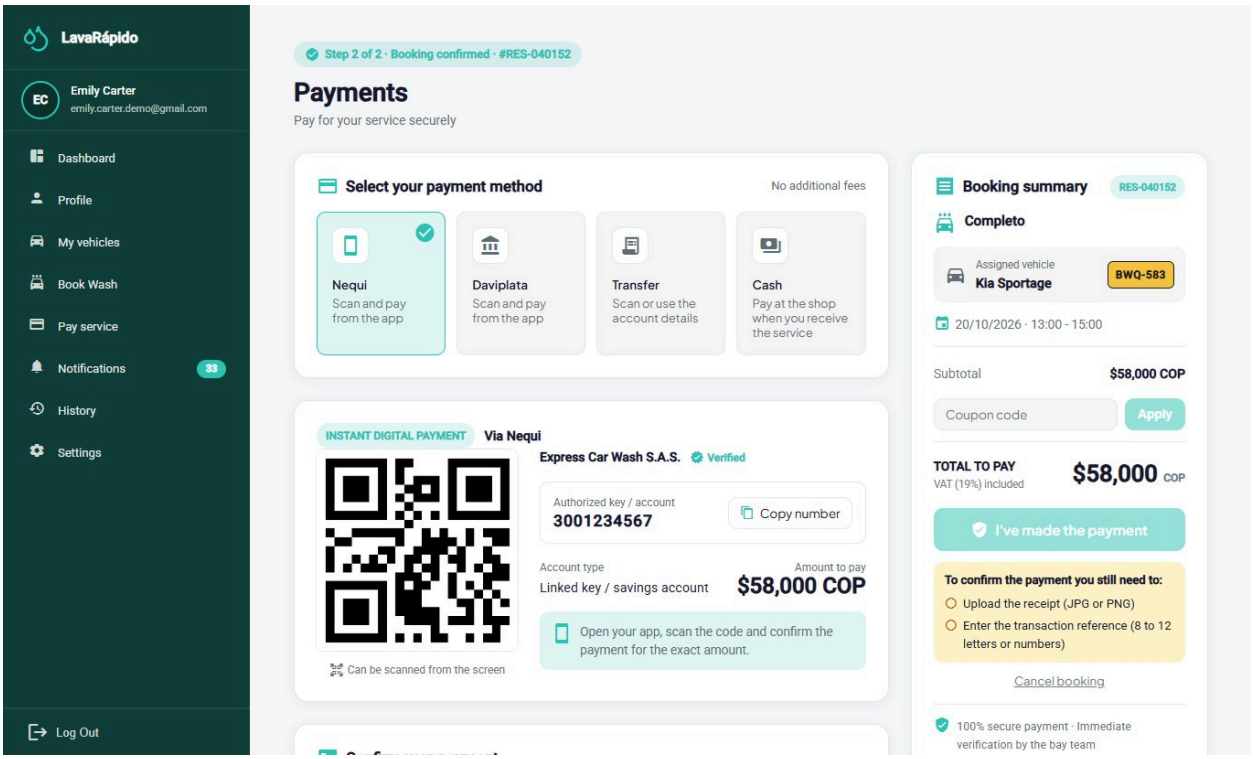


Figure 7

Customer Home (Mobile App)

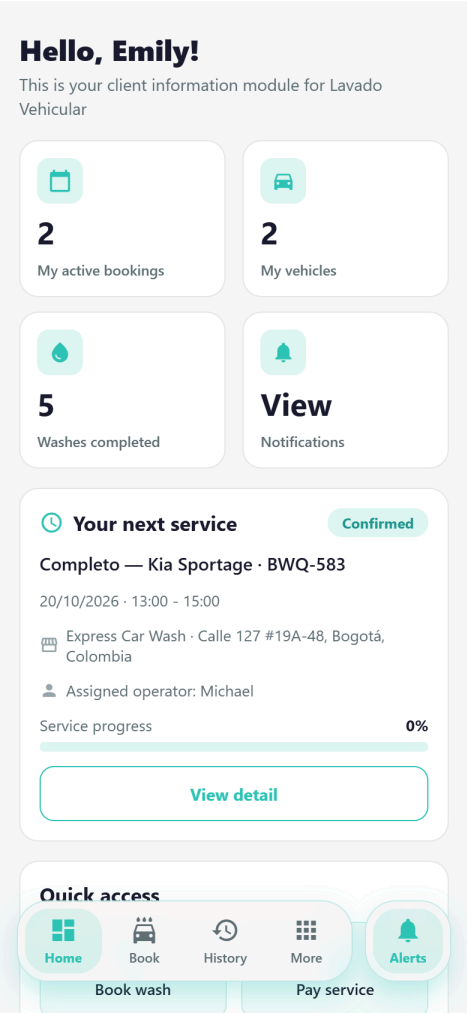


Figure 8

Assigned Services — Operator (Web)

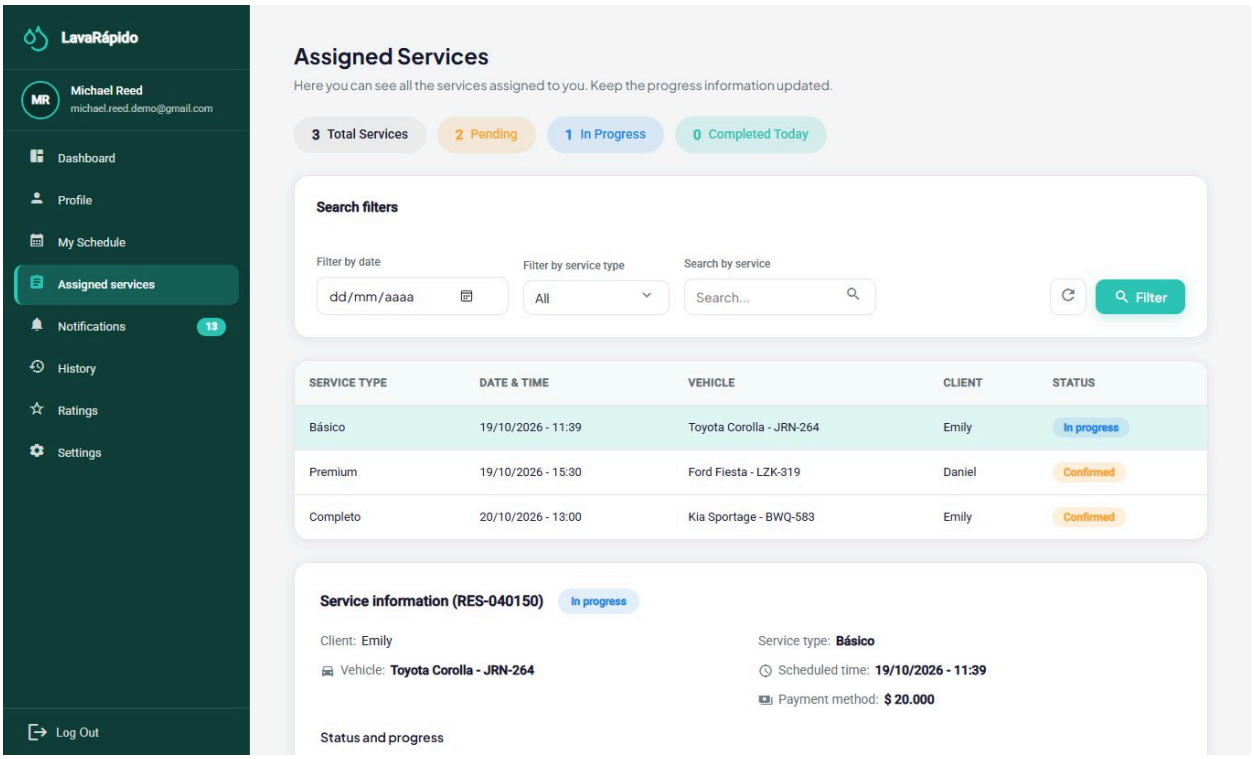


Figure 9

Operator Home (Mobile App)

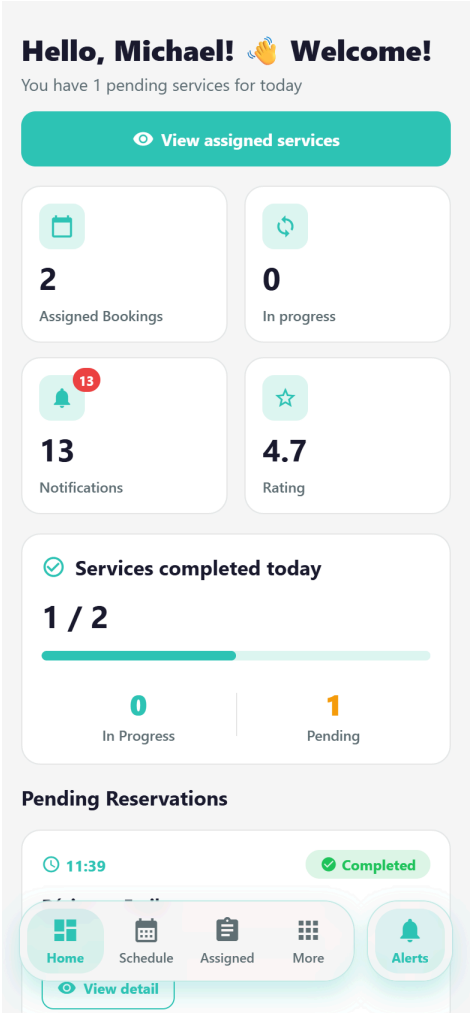


Figure 10

Dashboard — Administrator (Web)

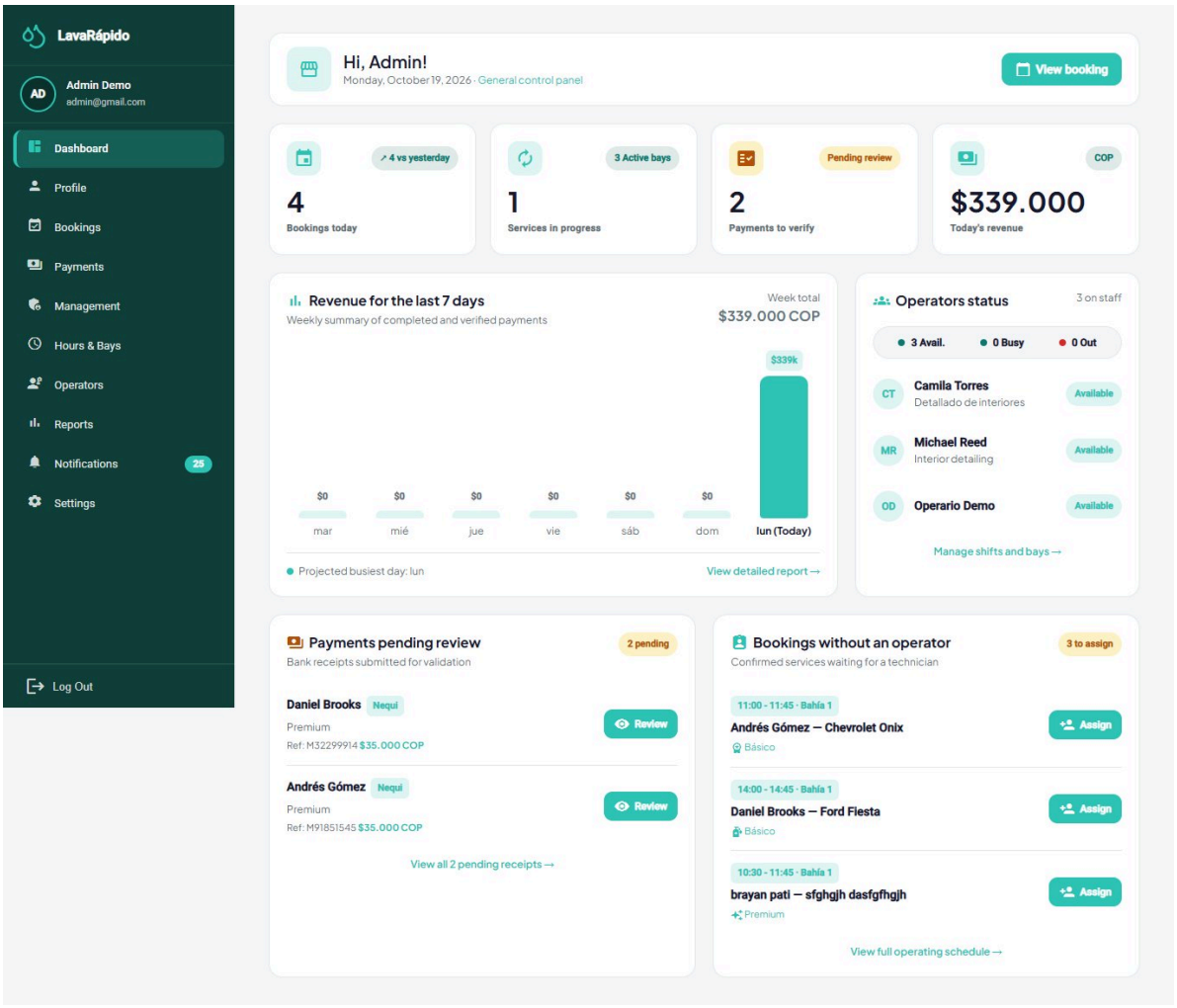


Figure 11

Booking Management — Administrator (Web)

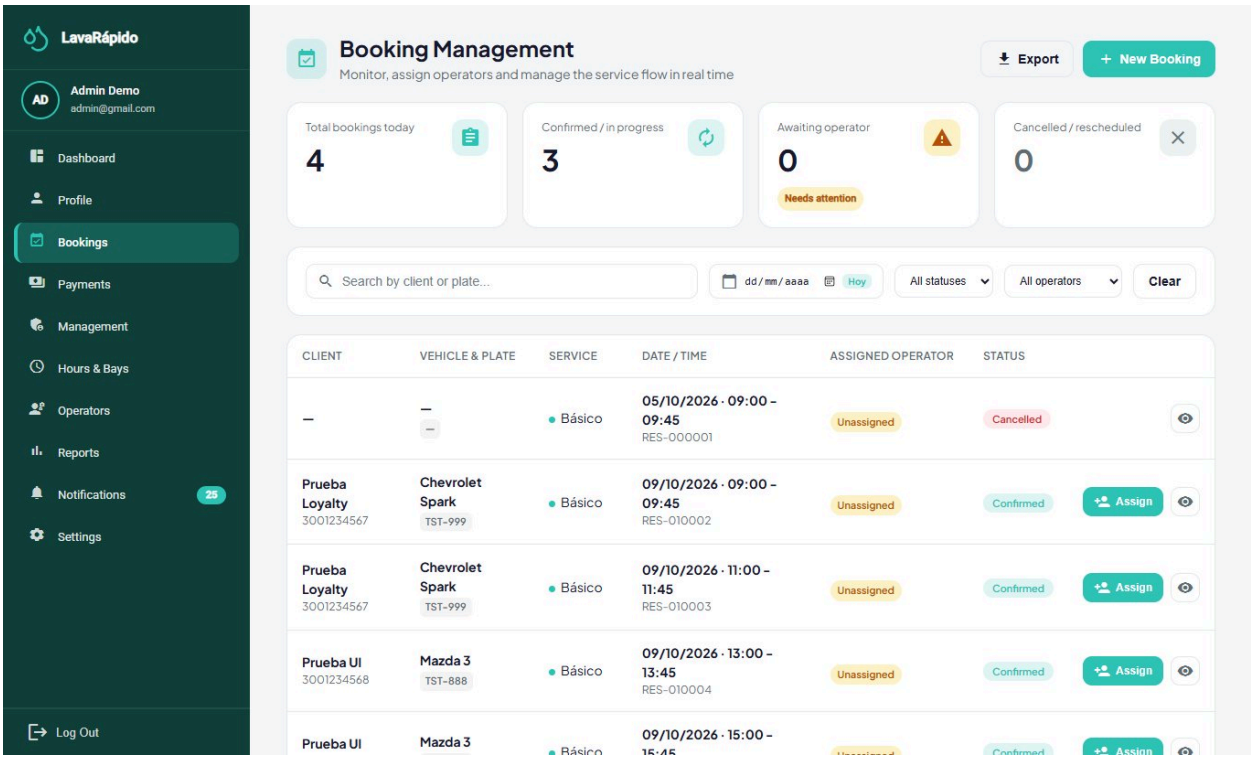
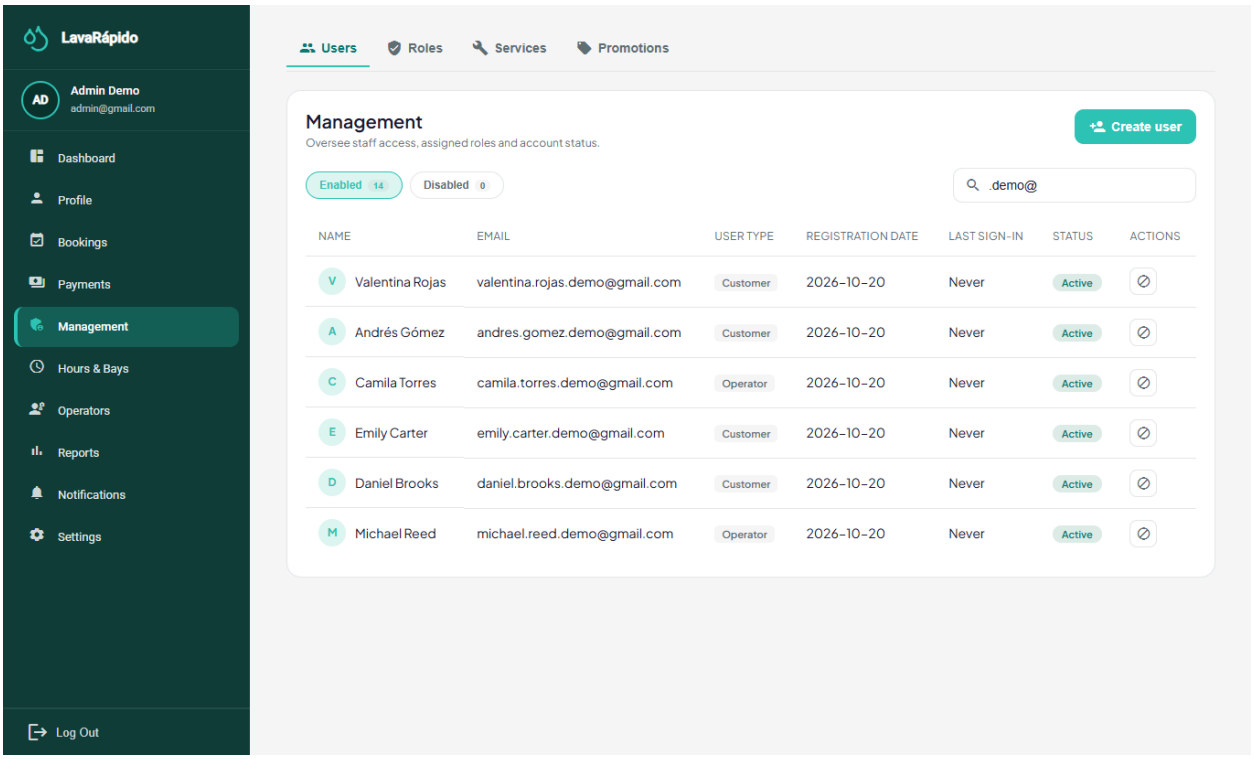


Figure 12

User, Role, Service, and Promotion Management — Administrator (Web)



> *Note.* Original screenshots taken with headless Chrome. The mobile app screenshots come from its web build; on Android the UI is the same, rendered with native components.

2.5 Responsive Design

Both applications are used from a width of 360 pixels (RNF-008), the narrowest Android phone the project supports, without horizontal scrolling.

- **Web.** The breakpoints are in `src/styles/_breakpoints.scss` and are applied with the `mobile` (up to 768 px) and `tablet` (up to 1024 px) mixins of `_mixins.scss`. From 768 px the side menu opens with the ☰ button, and the content of pages with a menu leaves 72 px at the top for that button (global rule in `styles.scss`). Headings use a relative line height of 1.25, because the Material theme sets 20 px on the `body` and a larger heading overlapped itself when it wrapped into two lines. Wide tables scroll inside their card; the operator's assigned services table is shown as cards from 600 px. On a phone, the customer dashboard places the stats and quick access two per row, the authentication screens use the full width, and the landing bar keeps only the language and **Login** from 420 px.
- **Mobile.** Screens are built with Flexbox: stat cards go two per row (`StatGrid`), button rows wrap, and filter inputs grow in width without losing their 46 px height.
- **Verification.** On October 17, 2026, 60 web screens (30 at 360 px and 30 at 768 px) and 26 mobile app screens at 360 px, in their web version, were measured with Playwright: none scrolls sideways or clips content. The team test `e2e/width-360.js` (lavarapido-infra) covers the same rule for the web.

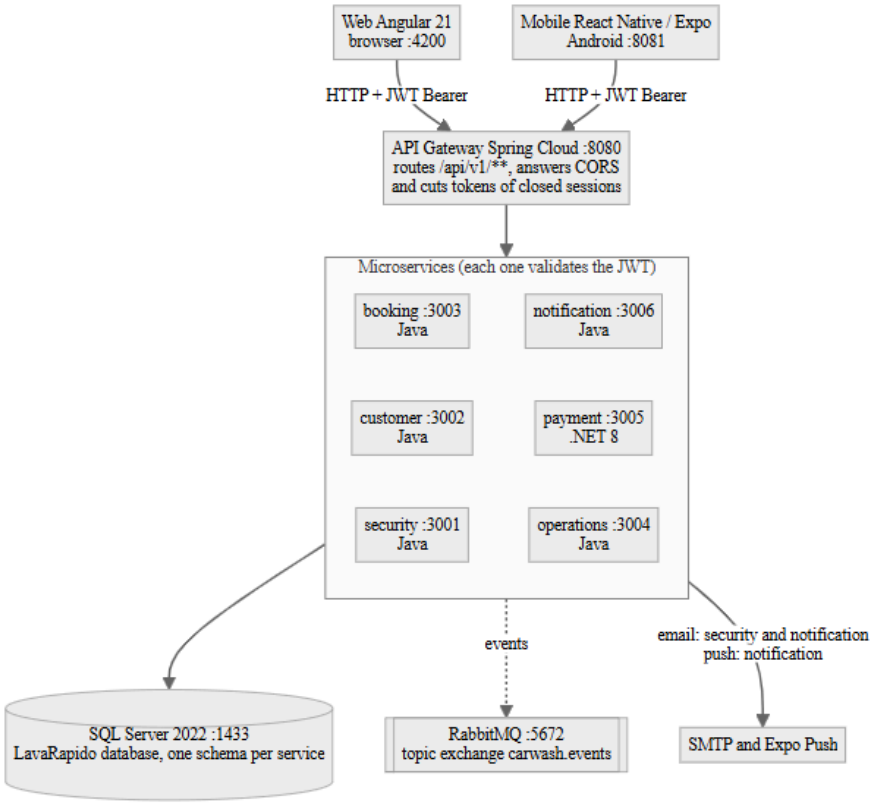
3. Architectural Model

The system follows a microservices architecture (Newman, 2021): an API Gateway as the single entry point, six domain services with hexagonal architecture, a SQL Server database split by schema, and RabbitMQ for asynchronous events.

3.1 Container Diagram (C4 Level 2)

Figure 13

System Container Diagram (C4, Level 2)



> *Note.* Notation based on the C4 model (Brown, n.d.). Clients talk only to the gateway; each service validates the JWT and stores data in its own schema. Since October 21, 2026 the gateway also rejects the tokens of closed sessions.

The web and mobile clients only call port 8080. The gateway does not check the JWT signature: it forwards the `Authorization` header and each service validates the token on its own (`Backend/lavarapido-api-gateway/src/main/resources/application.yml`, route comments, ADR-006). Since October 21, 2026 it does confirm that the session of the token is still open: `SessionCheckFilter`, which runs right after CORS, asks security-service with `GET /api/v1/auth/session` (204 if the session is still open, 401 if it was closed by a logout or because the account was deactivated) and keeps the answer for 30 seconds per SHA-256 fingerprint of the token, so an open session costs one extra call every half minute. If security-service does not answer within 3 seconds, the request goes through, and on logout the kept answer of that token is dropped. security-service makes the same check with the `sid` claim of the token (`ActiveSessionValidator`). It is configured with

SESSION_CHECK_ENABLED, SESSION_CHECK_CACHE_TTL, and SESSION_CHECK_TIMEOUT, which docker-compose.yml passes to the container since October 22, 2026 (SessionCheckFilter.java, SessionCheckConfig.java). If security-service does not answer within 3 seconds, the gateway lets the request through (*fail-open*). Since October 24, 2026 the filter checks every request with Authorization, with any scheme (bearer, Bearer, BEARER) and on any route, and the cache uses only the token, so logout clears it in all its variants (SEC-01, api-gateway 2e20204). That same day AuthRateLimitFilter (60 requests per minute per IP on /api/v1/auth/**, 429 RATE_LIMITED with Retry-After; AUTH_RATE_LIMIT_*), ClientIpResolver (it only trusts the X-Forwarded-For coming from TRUSTED_PROXIES, and forwards it to the services), and per-route timeouts (ROUTE_CONNECT_TIMEOUT 5 s and ROUTE_READ_TIMEOUT 60 s) were added. The service-to-service /internal/** routes have no gateway route and return 404 from outside.

3.2 Responsibility of Each Container

Table 5

Containers, Responsibility, and Owned Data

Container	Responsibility	Owned schemas	Public routes (gateway)
security-service	Sign-up, JWT login with lockout after failed attempts, password recovery, profile and address, preferences, users, roles, and permissions	security, audit	/auth/**, /users/**, /admin/users/**, /admin/roles-permissions/**
customer-service	Customers, vehicles, and vehicle types	customer	/vehicles/**, /vehicle-types/**, /admin/vehicles/**
booking-service	Catalog and prices, business hours, bays, availability, bookings, day summary, and business data	catalog, booking	/catalog/**, /schedule/**, /establishment, /bookings/**, /admin/bookings/**, /admin/catalog/**, /admin/schedule/**, /admin/bays/**, /admin/establishment, /operator/bookings/**
operations-service	Operators and their profile, availability, absences, automatic and manual assignment, execution, ratings, metrics, and inspection report	execution	/admin/operators/**, /operators/**, /admin/inspections/**, /public/inspections/**, /admin/assignments/**, /operator/services/**, /operator/ratings, /ratings/**
payment-service (.NET)	Payments with receipts, payment accounts, promotions, and loyalty points	payment, promotion	/payments/**, /payment-accounts, /admin/payments/**, /admin/payment-accounts/**, /admin/payment-methods, /admin/promotions/**, /loyalty/**
notification-service	Notification inbox, email, push (Expo), scheduled reminders, and alerts for bookings without an operator, in each account's language	notification	/notifications/**, /admin/notifications/**

Note. Source: routes in Backend/lavarapido-api-gateway/src/main/resources/application.yml and each service's Liquibase changelogs.

3.3 Service-to-Service Communication

There are two styles. Queries that need an immediate answer use synchronous REST; notices that something happened travel as events through RabbitMQ.

Table 6

Synchronous REST Calls Between Services

Source	Target	Purpose	Environment variable
booking	customer	Check that the vehicle belongs to the customer	`CUSTOMER_SERVICE_URL`
customer	security	Get the `person_id` if the sign-up event did not arrive	`SECURITY_SERVICE_URL`
operations	booking	Read, start, and finish bookings with the user's token; read bookings without a user for automatic assignment, free operators, and the follow-up job, and set to NO_SHOW a confirmed booking that ended without an operator (`/internal/v1/bookings`, `POST /internal/v1/bookings/{id}/no-show`)	`BOOKING_SERVICE_URL`, `BOOKING_INTERNAL_URL`, `INTERNAL_API_KEY`
operations	security	Operator and customer names, and whether the account is still active (internal route with a shared key)	`SECURITY_SERVICE_URL`, `SECURITY_INTERNAL_URL`, `INTERNAL_API_KEY`
payment	booking	Look up the booking and the amount due	`BOOKING_SERVICE_URL`
booking	payment	Before cancelling for lack of payment, find out whether the booking already has a payment in review or approved (`GET /internal/v1/payments/bookings/{bookingId}/reported`, answer `{"bookingId":123,"reported":true}`, 3 s timeout; since October 26, 2026). Without a key or with a wrong one, 401; a user's JWT does not work and the gateway does not route this path	`PAYMENT_SERVICE_URL` (`http://payment-service:3005`), `INTERNAL_API_KEY` (also in payment)
customer	booking	Know whether a vehicle has active bookings before removing it (`GET /internal/v1/bookings/active-count?customerVehicleId=`, since October 24, 2026)	`BOOKING_SERVICE_URL` (service root, without `/api/v1`), `INTERNAL_API_KEY`
notification	security	Email and language of each account and administrator list (`/internal/v1/users`)	`SECURITY_SERVICE_URL`, `INTERNAL_API_KEY`
gateway	security	Confirm that the session of the token is still open (`GET /api/v1/auth/session`; the answer is kept 30 seconds per token)	`ROUTE_SECURITY_URL`

Note. Source: `Backend/lavarapido-infra/docker-compose.yml` and, for the gateway, `SessionCheckConfig.java`.

Table 7

Events Published on the `carwash.events` Exchange

Routing key	Publisher	Consumer
`security.user_registered`	security	customer (creates the customer profile), notification (welcome)
`customer.provisioned`	customer (only written to the log)	None
`security.user.email-changed`, `security.user.password-changed`, `security.user.deactivated`	security (only written to the log)	None
`booking.confirmed`	booking	notification (notice and reminders), operations (automatic assignment, with the operator the customer chose)
`booking.cancelled`	booking (it carries `reason` and, since October 24, 2026, `reasonCode`)	notification
`booking.status_changed`	booking (when the administrator changes the status)	notification (notice to the customer; with IN_PROGRESS, COMPLETED, or NO_SHOW it cancels the reminders, and NO_SHOW uses the `BOOKING_NO_SHOW` type)
`execution.operator_assigned`, `execution.operator_unassigned` (since October 24, 2026), `execution.service_started`, `execution.service_completed`, `execution.inspection_shared`, `execution.service_not_started`, `execution.service_not_finished`	operations	notification
`payment.reported` (since October 26, 2026: the customer reported a payment with a receipt, also when reporting it again after a rejection; payload `paymentId`, `bookingId`, `customerUserId`, `amount`)	payment	booking (sets `payment_reported_at`, and the booking with digital payment no longer expires)
`payment.confirmed`	payment	notification; booking since October 26, 2026 (sets `payment_reported_at`)
`payment.rejected`, `payment.refunded`, `payment.promotion_redeemed`, `payment.loyalty_points_earned`	payment	notification

Note. Source: `BookingEvent.java:17-18`, `ExecutionEvent.java:14-19`, `InspectionSharedEvent.java:15`, `PaymentIntegrationEvents.cs`, `CustomerProvisioned.java:15`, and the `ROUTING_KEYS` list in `notification-service/.../RabbitMessagingConfig.java`. Since October 24, 2026 notification no longer listens to `booking.created`, which nobody published (H-12). Closing as NO_SHOW a confirmed booking that ended without an operator publishes no event, and neither does closing a booking as "Attended" from "To close": `ServiceCompleted` is not published, so the customer does not receive "ready for pickup" hours later. `ServiceNotStarted` and `ServiceNotFinished` become the `OPERATOR_SERVICE_NOT_REGISTERED` notice for the operator and `ADMIN_SERVICE_NOT_REGISTERED` for each active administrator (inbox and push, never email).

The notification and customer queues have their own dead-letter queue. Since October 21, 2026, a `null` field in an event payload no longer sends the whole message to notification's dead-letter queue: that field is ignored (`DomainEventEnvelope.java`). Since October 24, 2026 so does operations': it consumes from `operations-service.booking-confirmed` (only `booking.confirmed`), retries transient failures 3 times, and then sends the message to `operations-service.booking-confirmed.dlq`; the old `operations-service.events` queue is drained and deleted on its own at startup (`LegacyQueueMigration.java`), and operations publishes with broker confirmation. notification writes to the `pending_delivery`

table (changelog `047`; since changelog `048`, of October 24, 2026, `channel` and `language` are `NVARCHAR` like the rest of the text, because Hibernate reads it as `NCHAR` and the SQL Server driver does not convert `VARCHAR`), in the same transaction, the email and push it sends after the commit, and retries them (1, 2, 4… minutes, at most 1 hour, up to 5 attempts). notification stores the `event_id` of each event in `notification.processed_event` to ignore duplicates, because delivery is "at least once" (`DomainEventConsumerService.java:61-72`). Messaging is enabled with `MESSAGING_ENABLED`; when it is off, the Java services use a publisher that only writes to the log (`LoggingDomainEventPublisher.java:16`).

Since October 26, 2026 booking-service also consumes events (payment at booking, AC-88 and AC-89). Its queue `booking.payment-events` is bound to `carwash.events` with the keys `payment.reported` and `payment.confirmed`; a message that keeps failing after the retries goes to `booking.payment-events.dlq` through the `carwash.events.dlx` exchange, as in operations and notification (`RabbitConsumerConfig.java`). `PaymentEventsListener` reads `payload.bookingId` from the usual envelope (`{eventId, eventType, aggregateId, occurredAt, version, payload}`) and sets `payment_reported_at` (`Booking.markPaymentReported`, idempotent). `PaymentDeadlineJob` runs every `BOOKING_PAYMENT_EXPIRY_INTERVAL` (60 s) and is turned off with `BOOKING_PAYMENT_EXPIRY_ENABLED=false`: it looks for `DIGITAL` bookings without a report whose `payment_deadline` has passed and, **before cancelling** each one, asks payment-service through the internal path of Table 6 (outbound port `PaymentStatusDirectory`, adapter `PaymentServiceDirectory`). If `reported` is true, it marks the booking and does not cancel it (`PAYMENT_REPORTED`); if it is false, it cancels it through the usual path, with the reason `PAYMENT_TIMEOUT`, and publishes `BookingCancelled` (`CANCELLED`); if payment does not answer or returns an error, it does not cancel it in that round and checks it again in the next one (`PaymentTimeoutOutcome`). That query exists because there is no Outbox: in the live test of October 26 the event was lost and the job cancelled a paid booking (AC-89). The variables are `BOOKING_CASH_ACTIVE_LIMIT` (1), `BOOKING_PAYMENT_WINDOW` (30m), `BOOKING_PAYMENT_EXPIRY_ENABLED` (true), `BOOKING_PAYMENT_EXPIRY_INTERVAL` (60s), and `PAYMENT_SERVICE_URL`; payment-service now needs `INTERNAL_API_KEY`.

A scheduled job handles the bookings that ended and nobody closed, and the system only decides on its own when there is no doubt. Until October 23, 2026 booking-service also had `ExpiredBookingsJob`, which cancelled expired scheduled bookings; it was removed on October 24 (B-07) because every booking is created and rescheduled as confirmed and it never acted. In operations-service, `FollowUpJob` applies `FollowUpPolicy` every 5 minutes: a confirmed booking that never had an operator and ended more than an hour ago is set to NO_SHOW through booking's internal route, which checks the expiry again (409 `BOOKING_NOT_EXPIRED`); a confirmed booking with an operator that nobody started (`NOT_STARTED`) or an in-progress one that nobody finished one hour after its end (`NOT_FINISHED`) never changes on its own: it goes to the administrator's "To close" list (`GET /api/v1/admin/assignments/to-close`), who closes it as "Attended" (`POST /api/v1/admin/assignments/{bookingId}/attended`, 409 `BOOKING_NOT_TO_CLOSE` if it is not pending) or as "No-show" with the booking status change. To notify only once, `execution.service_execution.follow_up_notice_at` is stored (changelog `040`) and cleared when the service starts. This avoids recording as a no-show a service that was done and not registered. The variables are `BOOKING_EXPIRY_GRACE_MINUTES` (60, used by booking and operations), `BOOKING_EXPIRY_CHECK_ENABLED` and `BOOKING_EXPIRY_CHECK_INTERVAL` (booking, `true` and `PT5M`), and `FOLLOW_UP_ENABLED` and `FOLLOW_UP_INTERVAL` (operations, `true` and `PT5M`, it needs `INTERNAL_API_KEY`). The compose file passes them to the containers since October 22, 2026, including

`FOLLOW_UP_LOOK_BACK_DAYS` (28 days back, because the machine can stay off for weeks) (`FollowUpJob.java`, `FollowUpService.java`).

3.4 Internal Layers of Each Service (Hexagonal)

Figure 14

Layers of a Service With Hexagonal Architecture

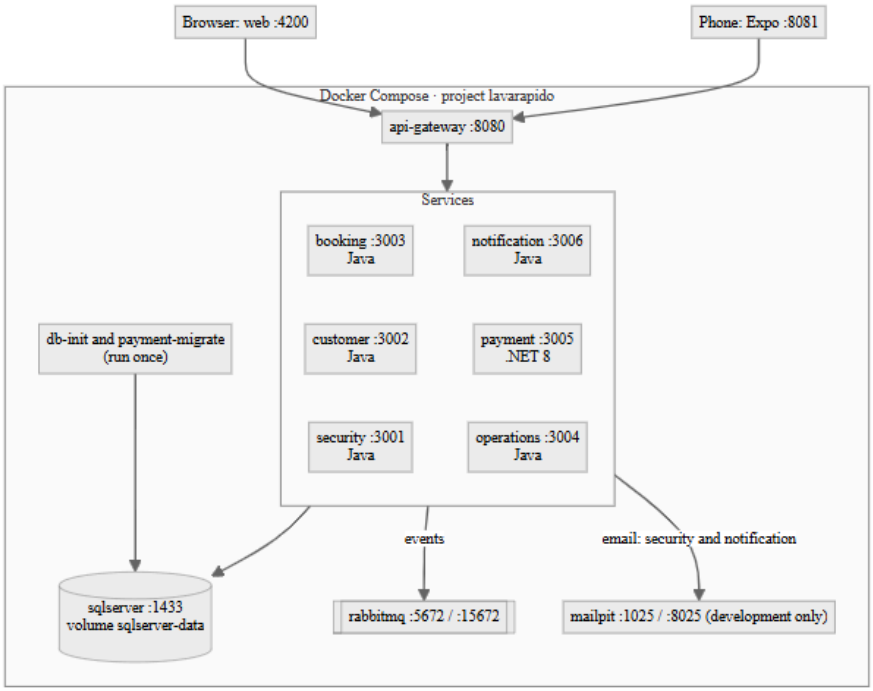


Hexagonal architecture separates the domain core from infrastructure through ports and adapters (Cockburn, 2005); domain modeling follows Evans (2004). The Java services use the `domain`, `application`, and `infrastructure` packages (for example, `security-service/src/main/java/com/lavarapido/security/`). payment-service applies the same idea with four .NET projects: `PaymentService.Domain`, `.Application` (with `Ports/In` and `Ports/Out`), `.Infrastructure`, and `.Api`.

3.5 Deployment

Figure 15

Deployment Diagram With Docker Compose



`docker compose up -d` starts only SQL Server, Mailpit, and RabbitMQ, and the services run from the IDE. `docker compose --profile app up -d --build` also starts the seven services in containers (`Backend/lavarapido-infra/docker-compose.yml:1-`

6). The Java services create their tables with Liquibase on startup; payment-service creates them with the `payment-migrate` container.

In production the same compose file runs on an Azure virtual machine with `docker-compose.prod.yml` on top: two more containers are added, `web` (nginx with the Angular build) and `caddy`, which is the only one with published ports (80 and 443, HTTPS certificate managed by Caddy). Caddy sends `/api/*` to the gateway, `/app/*` to the APK download and its `version.json`, and everything else to the web. No service publishes ports on the host, Mailpit is not started (emails go out through Gmail SMTP), and the server is updated with `deploy/update-server.sh`, which deploys the `main` branch of every repository since infra *v1.1.1* (October 26, 2026; before, it deployed `develop` with the `BRANCH` variable), from the GitHub Actions workflow "Desplegar en Azure" (`deploy/ci-deploy.sh`) or by a team member over SSH. Since October 24 `update-server.sh` keeps the running version as `:prev` before rebuilding and waits for every container to be healthy, `rollback.sh` goes back to `:prev` without building, and `setup-server.sh` never downgrades. The `web` container runs `nginx-unprivileged`, without `root`, on port 8080, and Caddy points to `web:8080`. Continuous deployment (every approved merge into `main` deploys on its own, and when the machine is turned on `deploy/boot-update.sh` deploys what came in while it was off) runs on its own since October 26, 2026, with the "Pedir despliegue" job in 8 repositories and the mobile OTA (`cd.yml`), which only publishes if the native fingerprint matches the published APK's (Deployment Document, sections 1.3.12 and 4.3.4). Since October 20 and 21, 2026: Caddy sends security headers and a content security policy (CSP); every container has a health check every 30 seconds and its logs are capped at 3 files of 10 MB; the database is backed up every day and before every update (`deploy/backup-db.sh`), each backup is checked by restoring it under another name (`deploy/verify-backup.sh`), and the last 14 are kept (`deploy/restore-db.sh` restores one). Source: `lavarapido-infra/docker-compose.prod.yml`, `deploy/Caddyfile`, `deploy/backup-db.sh`, `.github/workflows/deploy.yml`; diagram in `vehicle-w-docs/08-uml/diagrams/source/deployment-production.md`.

4. Design Patterns

The code applies 18 verifiable patterns: 5 architectural, 8 backend design patterns, and 5 in the clients. The code itself names some of them in its comments (Strategy, Factory, Unit of Work). Pattern names follow the catalogs of Gamma et al. (1994), Fowler (2002), and Hohpe and Woolf (2003).

Table 8

Design Patterns Identified

#	Pattern	Where it is used	Problem it solves
1	Microservices + API Gateway	`Backend/lavarapido-api-gateway/.../application.yml`	A single entry point for web and mobile; each domain is deployed separately
2	Hexagonal architecture (ports and adapters)	`domain/port/in`, `domain/port/out`, `infrastructure/adapter` packages in each Java service; `Ports/In`, `Ports/Out` in payment	The domain does not depend on Spring, JPA, or RabbitMQ; infrastructure can change without touching the rules
3	Schema per service in a single database (a variant of "database per service")	A single `LavaRapido` database with one schema per service; Liquibase changelogs per service; no FKs across schemas. All services connect with the same `sa` account, so the separation is by convention and not by permissions (H-10)	No service reads another's tables; cross references are validated through the API
4	Publish / subscribe (distributed Observer)	`RabbitDomainEventPublisher.java`, `DomainEventListener.java:38`, `UserRegisteredListener.java:38`	Notify other services without coupling them or waiting for a reply
5	Idempotent consumer	`DomainEventConsumerService.java:61-72` + table `notification.processed_event`	RabbitMQ delivers "at least once"; a repeated event does not create two notifications
6	Repository	`UserAccountRepository` (port) → `UserAccountPersistenceAdapter.java:33` → `AppUserJpaRepository.java:12`	Hide data access behind a domain interface
7	Adapter	`BCryptPasswordHasher.java:10` implements `PasswordHasher`; `HttpAuthService` implements `IAuthService` on mobile	Adapt an external library (BCrypt, fetch) to the contract the domain expects
8	Strategy	`IDiscountStrategy.cs`, `PercentageDiscountStrategy.cs`, `FixedAmountDiscountStrategy.cs`	Each discount type computes in its own way, with no `switch` inside `Promotion`
9	Factory	`DiscountStrategyFactory.cs:9-21`	A single place decides which strategy matches each `DiscountType`
10	Unit of Work	`IUnitOfWork.cs:8-15`, `EfUnitOfWork.cs`	Approving a payment and crediting points are saved together or not at all
11	Value Object	`EmailAddress.java:12` (record validated on construction), `DocumentNumber`, `PersonName`, `HashedPassword`	Invalid data cannot exist inside the domain
12	Aggregate Root + domain events	`AggregateRoot.cs:24-39` (payment), `domain/event/*` (Java)	The root entity protects its rules and records what happened
13	DTO	`LoginRequest.java:6-8` (record with validation), `dto/*` in each service; `Contracts/*` in payment	Separate the HTTP contract from the domain model
14	Dependency injection	`DomainConfig.java` (`@Bean`), `DependencyInjection.cs` (.NET), `inject()` in Angular, `AuthServiceProvider` on mobile	Objects receive their dependencies; tests pass in doubles
15	Null Object / alternate implementation by configuration	`LoggingDomainEventPublisher.java:16` and `NullIntegrationEventPublisher.cs`	With `MESSAGING_ENABLED=false` the system works without RabbitMQ
16	Guard + Interceptor (chain of responsibility)	`Web/src/app/core/guards/auth-guard.ts:10-34`, `core/interceptors/auth-interceptor.ts:14-38`	Control access by role and attach the token only to the system's own API

#	Pattern	Where it is used	Problem it solves
17	Domain policy (Specification)	`AssignmentPolicy.java`, `AutoAssignmentPolicy.java`, `StartPolicy.java` (operations); `LoginLockout.java` (security)	Each business rule (who can take a booking, who gets it automatically, when it can start, when an account is locked) lives in a pure domain class that is tested without a database
18	MVVM (Model-View-ViewModel)	Web: `features/*/viewmodels/*.viewmodel.ts` and `shared/viewmodels/*.viewmodel.ts` (36 `@Injectable` ViewModels with signals, for example `payment.viewmodel.ts` and `reservations.viewmodel.ts`); mobile: `modules/*/viewmodels/use*.ts` hooks (20, for example `useInspection.ts` and `useSettings.ts`)	The page or screen only displays and forwards events; the state, validations, and service calls live in the ViewModel, which is tested without the view

> *Note.* Other mechanisms present: the `X-Correlation-Id` correlation filter (`CorrelationIdFilter.java:23-39`), the gateway session filter with the answer kept 30 seconds per token (`SessionCheckFilter.java`), the scheduled follow-up job for bookings nobody closed (`FollowUpJob.java`; `ExpiredBookingsJob.java` was removed on October 24, 2026), the gateway request limit (`AuthRateLimitFilter.java`), the web's automatic refresh (`LiveRefresh`/`keepFresh`, since October 24, 2026), Angular signal-based stores for admin state (`features/admin/services/*-store.ts`), and lazy loading of modules per role (`loadChildren` in `app.routes.ts`).

Client architecture. The web follows MVVM: each page in `features/<area>/pages` (the view, a *standalone* Angular component) injects its ViewModel from `features/<area>/viewmodels` or, if several areas share it (profile, settings, create user, sidebar), from `shared/viewmodels`; the ViewModel exposes signals (`signal`, `computed`) and methods, and calls the HTTP services in `core/services` (the Model). This way each screen's rules can be tested with Vitest without mounting the view; as of October 23, 2026, 9 of the 36 ViewModels have their `*.viewmodel.spec.ts`. The mobile app applies the same idea with React hooks and is organized by layers: `modules/<role>/screens` (view), `modules/<role>/viewmodels/use*.ts` (state and actions), `core/services` (HTTP clients for the gateway), `core/api` (`fetch` client and error translation), `core/navigation` (each role only registers its screens) and `core/storage` (session in secure storage), `shared` (components, validators, and i18n), and `app` (theme, providers, and configuration). Ten mobile screens or components still call the services without a ViewModel (for example `ClientPaymentScreen.tsx`). App updates use two paths (`app/providers/AppUpdatesCheck.tsx`): a new APK, mandatory when something native changes, and EAS Update (`expo-updates`, `runtimeVersion` with the *fingerprint* policy, *preview* channel, check when the app opens) for changes to screens and logic only (`npm run publish:update`).

4.1 Representative Snippets

Strategy + Factory (`Backend/lavarapido-payment-service/src/PaymentService.Domain/Promotions/Discounts/DiscountStrategyFactory.cs:9-21`):

```
public static class DiscountStrategyFactory
{
    private static readonly IDiscountStrategy Percentage = new PercentageDiscountStrategy();
    private static readonly IDiscountStrategy FixedAmount = new FixedAmountDiscountStrategy();

    public static IDiscountStrategy For(DiscountType type) => type switch
    {
        DiscountType.Percentage => Percentage,
        DiscountType.FixedAmount => FixedAmount,
        _ => throw new DomainException(DomainErrorCodes.PromotionDiscountUnsupported, "...")
    };
}
```

Outbound port + Adapter (`security-service/.../domain/port/out/PasswordHasher.java` and `BCryptPasswordHasher.java`):

```
public interface PasswordHasher {
    HashedPassword hash(String raw);
    boolean matches(String raw, HashedPassword hash);
}

class BCryptPasswordHasher implements PasswordHasher {
    private final PasswordEncoder encoder;
    public boolean matches(String raw, HashedPassword hash) {
        return raw != null && encoder.matches(raw, hash.value());
    }
}
```

Alternate implementation by configuration (`LoggingDomainEventPublisher.java:16-17`):

```
@ConditionalOnProperty(prefix = "app.messaging", name = "enabled", havingValue = "false", matchIfMissing = true)
class LoggingDomainEventPublisher implements DomainEventPublisher { ... }
```

HTTP interceptor (`Web/src/app/core/interceptors/auth-interceptor.ts:14-25`):

```
export const authInterceptor: HttpInterceptorFn = (req, next) => {
  if (!req.url.startsWith(API_BASE_URL)) {
    return next(req);
  }
  const token = inject(AuthService).accessToken();
  const request = token
    ? req.clone({ setHeaders: { Authorization: `Bearer ${token}` } })
    : req;
  ...
};
```

5. Data Model

The `LavaRapido` database (SQL Server 2022) has 58 tables in 9 schemas, created by each service's Liquibase changelogs. Foreign keys only exist within a single schema; references to another service are indexed columns without an FK, validated through the API.

Table 9

Schemas, Owner, and Number of Tables

Schema	Owning service	Tables	Changelog
`security`	security-service	11	`001`–`039` in `security-service/src/main/resources/db/changelog`
`audit`	security-service	1	`002-create-security-and-audit-core.sql`
`customer`	customer-service	3	`004`, `014`, `015`
`catalog`	booking-service	3	`005`, `022`
`booking`	booking-service	10	`005`, `018`, `021`, `024`, `032`
`execution`	operations-service	11	`007`, `033`, `037`
`payment`	payment-service	7	`lavarapido-payment-service/db/changelog/changes/008`–`023`
`promotion`	payment-service	5	`003`–`007`, `017`, `018`
`notification`	notification-service	7	`010`–`048`

> *Note.* The `security.custom_role` table was created in `023` and dropped in `026`; it is not counted. Earlier documents (`MODELO-DE-DATOS.md`) mention 43 tables: the changelogs later added `bay_status`, `schedule_history`, `device_token`, `processed_event`, `booking_reminder`, `booking_tracking`, and, on October 24, 2026, `pending_delivery` (notification changelog `047`). The 58 tables were counted in the real database on October 27, 2026.

5.1 Common Conventions

- **Audit block.** Almost every domain table ends with `created_at`, `created_by`, `updated_at`, `updated_by`, `deleted_at`, `deleted_by`, and `row_version`. It is absent from the base `security` tables (`person`, `app_user`, `role`, `permission`, `role_permission`, `user_role`, `user_session`, `password_reset_token`), `audit.audit_log`, `booking.schedule_history`, and `notification.booking_tracking`. The tables added in October (`security.login_lockout`, `security.person_address`, `execution.operator_profile`, `execution.operator_skill`, and the four inspection report tables) do not have the full block either: they only keep the date and author columns they use.
- **Soft delete.** A non-null `deleted_at` marks a deleted row; several unique indexes are filtered with `WHERE deleted_at IS NULL`.
- **Optimistic concurrency.** A `tr_touch_<table>` trigger updates `updated_at` and adds 1 to `row_version` on every UPDATE of the tables that are edited (16 triggers); in payment, `EfUnitOfWork` does it before saving. Only payment-service compares `row_version` when updating (EF Core concurrency token); the Java services do not check it yet.
- **Application locks.** Where two operations at the same time could clash, the Java services take a SQL Server lock (`sp_getapplock`) that is released when the transaction ends: security-service uses `login-lockout-{userId}` at sign-in, so that two simultaneous failed attempts on the same account do not overwrite each other's lockout count (`LoginLockoutPersistenceAdapter.lock`); booking-service uses `booking-schedule` when it creates or reschedules a booking and when it closes the expired ones, and operations-service uses `operations-assignment` (since October 21, 2026) in manual and automatic assignment, so an operator does not end up with two services at the same time (`BookingPersistenceAdapter.lockSchedule`, `ExecutionPersistenceAdapter.lockAssignments`).
- **UTC dates.** All default values use `SYSUTCDATETIME()`.
- **Coded catalogs.** States and types are tables with a unique `code` (`booking_status`, `payment_status`, `execution_status`, etc.) loaded by seed changelogs.

5.2 Entity-Relationship Model

Solid lines are real FKs; dashed lines are logical references to another service (no FK). Only key attributes are shown; full detail is in the data dictionary (5.4).

Figure 16

Security and Customers (`security`, `audit`, `customer`)

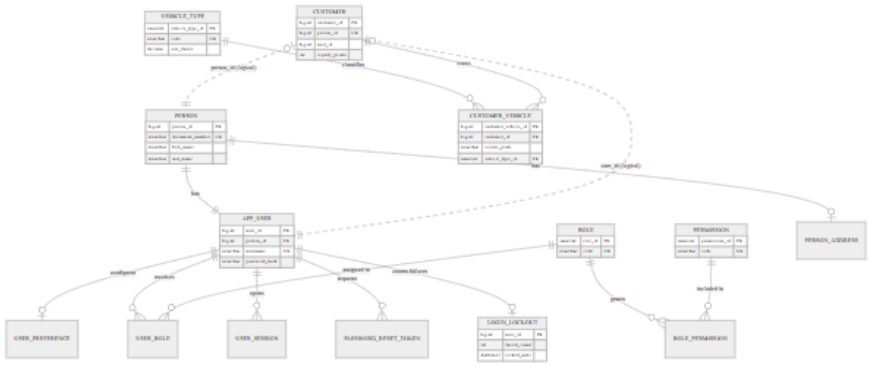
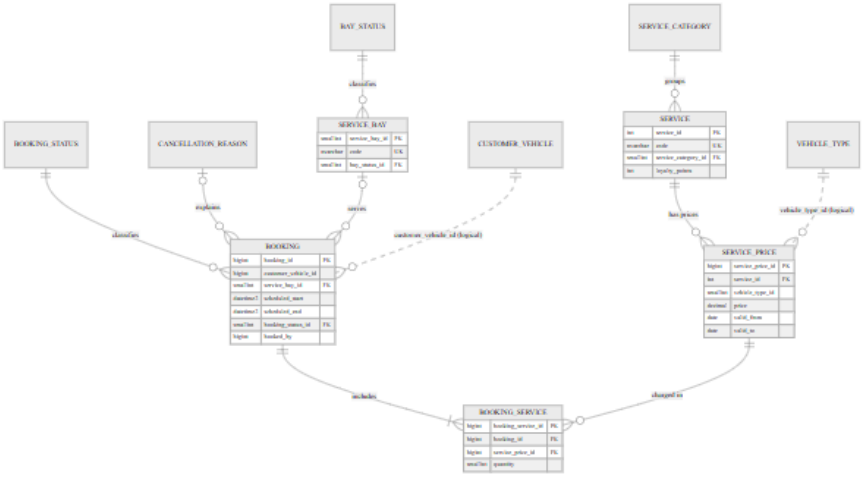


Figure 17

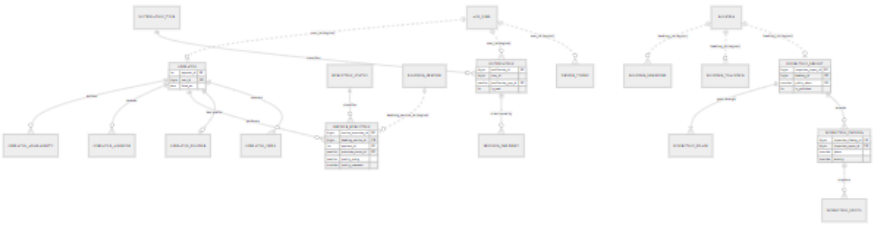
Catalog and Bookings (`catalog`, `booking`)



Standalone tables in the `booking` schema (no relationships): `establishment` (single row, `singleton = 1`), `business_hour` (one row per day), `business_hour_exception` (one per date), and `schedule_history`.

Figure 18

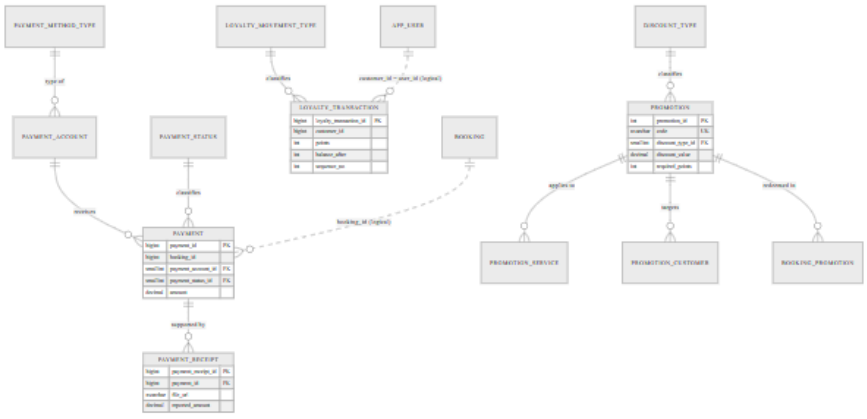
Execution and Notifications (`execution`, `notification`)



`notification.processed_event` has no relationships: it stores the `event_id` of each event already handled.

Figure 19

Payments, Promotions, and Loyalty (`payment`, `promotion`)



5.3 Relational Model

Notation: PK = primary key, UK = unique value, → = foreign key, ⇢ = logical reference to another service (no FK). The audit block is omitted.

- `security.person(person_id PK, document_number UK, first_name, last_name, phone, email)`
- `security.app_user(user_id PK, person_id → person UK, username UK, password_hash, is_active, last_login, created_at, terms_accepted_at, data_policy_accepted_at)`
- `security.login_lockout(user_id PK → app_user, failed_count, first_failed_at, locked_until, updated_at)` · `security.person_address(person_id PK → person, address, updated_at)`
- `security.role(role_id PK, code UK, name)` · `security.permission(permission_id PK, code UK, name, resource, action)`
- `security.role_permission(role_permission_id PK, role_id → role, permission_id → permission)` · `security.user_role(user_role_id PK, user_id → app_user, role_id → role)`

- security.user_session(user_session_id PK, user_id → app_user, refresh_token_hash, started_at, expires_at, revoked_at, ip_address, user_agent)
- security.password_reset_token(password_reset_token_id PK, user_id → app_user, token_hash UK, requested_at, expires_at, used_at, ip_address)
- security.user_preference(user_preference_id PK, user_id → app_user UK, theme, language, notifications_enabled, email_reminders_enabled, promotions_enabled)
- audit.audit_log(audit_log_id PK, entity, entity_id, action, previous_data, new_data, changed_by, changed_at)
- customer.vehicle_type(vehicle_type_id PK, code UK, name, size_factor, display_order, is_active)
- customer.customer(customer_id PK, person_id ⇢ security.person UK, user_id ⇢ security.app_user, loyalty_points, customer_since)
- customer.customer_vehicle(customer_vehicle_id PK, customer_id → customer, license_plate, vehicle_type_id → vehicle_type, brand, model, color)
- catalog.service_category(service_category_id PK, code UK, name, description, display_order, is_active)
- catalog.service(service_id PK, code UK, name, description, service_category_id → service_category, loyalty_points, is_active)
- catalog.service_price(service_price_id PK, service_id → service, vehicle_type_id ⇢ customer.vehicle_type, price, estimated_minutes, valid_from, valid_to)
- booking.establishment(establishment_id PK, legal_name, trade_name, tax_id, address, phone, email, logo_url, singleton UK, business_type, founded_on, legal_representative, legal_representative_document, website, tax_regime, ciiu_activity, dian_resolution, invoice_prefix, invoice_range, instagram, facebook, support_line, service_hours_note)
- booking.business_hour(business_hour_id PK, day_of_week UK, opens_at, closes_at, break_starts_at, break_ends_at, is_active)
- booking.business_hour_exception(business_hour_exception_id PK, exception_date UK, is_closed, opens_at, closes_at, reason)
- booking.bay_status(bay_status_id PK, code UK, name, accepts_bookings, display_order) · booking.service_bay(service_bay_id PK, code UK, name, bay_status_id → bay_status)
- booking.booking_status(booking_status_id PK, code UK, name, is_final, display_order, is_active) · booking.cancellation_reason(cancellation_reason_id PK, code UK, name, is_customer_fault, display_order, is_active)
- booking.booking(booking_id PK, customer_vehicle_id ⇢ customer.customer_vehicle, service_bay_id → service_bay, scheduled_start, scheduled_end, booking_status_id → booking_status, booked_by ⇢ security.app_user, cancellation_reason_id → cancellation_reason, points_redeemed, points_discount_amount, notes, payment_mode, payment_deadline, payment_reported_at)

- booking.booking_service(booking_service_id PK, booking_id → booking, service_price_id → catalog.service_price, quantity)
- booking.schedule_history(schedule_history_id PK, entity_type, title, detail, changed_at, changed_by)
- execution.execution_status(execution_status_id PK, code UK, name, is_final, display_order, is_active)
- execution.operator(operator_id PK, user_id ⇢ security.app_user UK, hired_on, is_active)
- execution.operator_availability(operator_availability_id PK, operator_id → operator, day_of_week, starts_at, ends_at, is_active)
- execution.operator_absence(operator_absence_id PK, operator_id → operator, starts_at, ends_at, reason)
- execution.service_execution(service_execution_id PK, booking_service_id ⇢ booking.booking_service UK, operator_id → operator, execution_status_id → execution_status, started_at, finished_at, quality_rating, quality_comment, rated_at, is_comment_visible, follow_up_notice_at)
- execution.operator_profile(operator_id PK → operator, specialty, updated_at, updated_by) · execution.operator_skill(operator_skill_id PK, operator_id → operator, name, skill_level, display_order)
- execution.inspection_report(inspection_report_id PK, booking_id ⇢ booking.booking UK, booking_code, vehicle, plate, services, booking_date, booking_time, duration_minutes, public_token UK, is_published, published_at)
- execution.inspection_phase(inspection_phase_id PK, inspection_report_id → inspection_report, phase, status, started_at, finished_at) · execution.inspection_finding(inspection_finding_id PK, inspection_report_id → inspection_report, phase, area, severity, note) · execution.inspection_photo(inspection_photo_id PK, inspection_finding_id → inspection_finding, content_type, content, size_bytes)
- payment.payment_method_type(payment_method_type_id PK, code UK, name, requires_account, requires_receipt, display_order, is_active)
- payment.payment_account(payment_account_id PK, payment_method_type_id → payment_method_type, account_holder, account_number, qr_image_url, instructions, is_active)
- payment.payment_status(payment_status_id PK, code UK, name, is_final, display_order, is_active)
- payment.payment(payment_id PK, booking_id ⇢ booking.booking, payment_account_id → payment_account, payment_status_id → payment_status, amount, processed_at, approved_by, rejection_reason)
- payment.payment_receipt(payment_receipt_id PK, payment_id → payment, file_url, transaction_reference, reported_amount, uploaded_at, uploaded_by, reviewed_at, reviewed_by, review_comment)
- payment.loyalty_movement_type(loyalty_movement_type_id PK, code UK, name, sign, display_order, is_active)
- payment.loyalty_transaction(loyalty_transaction_id PK, customer_id ⇢ security.app_user (holds the customer's user_id, ADR-015), loyalty_movement_type_id → loyalty_movement_type, booking_id ⇢

booking.booking, points, balance_after, sequence_no, expires_on, description)

- promotion.discount_type(discount_type_id PK, code UK, name, display_order, is_active)
- promotion.promotion(promotion_id PK, code UK, name, description, discount_type_id → discount_type, discount_value, max_discount_amount, min_purchase_amount, is_public, min_completed_booking, valid_from, valid_to, max_redemptions, max_redemptions_per_customer, is_active, price, duration_minutes, icon, featured, benefits, required_points)
- promotion.promotion_service(promotion_service_id PK, promotion_id → promotion, service_id ⇢ catalog.service)
- promotion.promotion_customer(promotion_customer_id PK, promotion_id → promotion, customer_id ⇢ customer.customer, assigned_at, assigned_by)
- promotion.booking_promotion(booking_promotion_id PK, booking_id ⇢ booking.booking, promotion_id → promotion, applied_amount, discount_percent, max_discount_amount)
- notification.notification_type(notification_type_id PK, code UK, name, display_order, is_active)
- notification.notification(notification_id PK, user_id ⇢ security.app_user, notification_type_id → notification_type, title, message, reference_entity, reference_id, is_read, read_at, sent_at)
- notification.device_token(device_token_id PK, user_id ⇢ security.app_user, token UK, platform, is_active, last_seen_at)
- notification.processed_event(processed_event_id PK, event_id UK, event_type, processed_at)
- notification.booking_reminder(booking_reminder_id PK, booking_id ⇢ booking.booking, user_id ⇢ security.app_user, booking_code, scheduled_start, lead_minutes, remind_at, sent_at)
- notification.booking_tracking(booking_id PK ⇢ booking.booking, booking_code, customer_user_id, operator_user_id, scheduled_start, is_cancelled, unassigned_alert_at, updated_at)
- notification.pending_delivery(pending_delivery_id PK, notification_id → notification, channel, email_address, recipient_name, language, attempts, next_attempt_at, last_error, delivered_at, failed_at, created_at), with UK (notification_id, channel)

5.4 Data Dictionary

There is one dictionary for each of the 58 tables, grouped by schema. The "Null" column shows whether the field accepts NULL; PK = primary key, FK = foreign key, UK = unique value, and ⇢ = logical reference to another service (no FK). Tables that carry the audit block (section 5.1) show it as a single row at the end.

Schema `security`

Accounts, roles, permissions, sessions, and preferences. Owner: security-service.

Table 10

Data Dictionary of `security.person` — Personal Data

Field	Type	Null	Key	Description
person_id	BIGINT IDENTITY	No	PK	Person identifier
document_number	NVARCHAR(20)	No	UK	ID document number
first_name	NVARCHAR(60)	No		First names
last_name	NVARCHAR(60)	No		Last names
phone	NVARCHAR(20)	Yes		Phone
email	NVARCHAR(120)	Yes		Contact email

Table 11

Data Dictionary of `security.app_user` — Login Account

Field	Type	Null	Key	Description
user_id	BIGINT IDENTITY	No	PK	User identifier
person_id	BIGINT	No	FK → person, UK	Person who owns the account (1 to 1)
username	NVARCHAR(60)	No	UK	Email used to log in
password_hash	NVARCHAR(255)	No		BCrypt password hash
is_active	BIT	No		Account enabled (default 1)
last_login	DATETIME2(3)	Yes		Last login
created_at	DATETIME2(3)	No		Account registration date (migration `036`)
terms_accepted_at	DATETIME2(3)	Yes		When the person accepted the Terms at sign-up (migration `039`)
data_policy_accepted_at	DATETIME2(3)	Yes		When the person accepted the Data Processing Policy at sign-up (migration `039`, Law 1581 of 2012)

Table 12

Data Dictionary of `security.role` — System Roles

Field	Type	Null	Key	Description
role_id	SMALLINT IDENTITY	No	PK	Role identifier
code	NVARCHAR(30)	No	UK	`ADMIN`, `OPERATOR`, or `CLIENT` (seed `101`)
name	NVARCHAR(60)	No		Display name

Table 13

Data Dictionary of `security.permission` — Permissions

Field	Type	Null	Key	Description
permission_id	SMALLINT IDENTITY	No	PK	Permission identifier
code	NVARCHAR(60)	No	UK	`VIEW_PANELS`, `CREATE_RECORDS`, `EDIT_DATA`, or `DELETE` (changelog `026`)
name	NVARCHAR(80)	No		Display name
resource	NVARCHAR(60)	No		Resource it applies to
action	NVARCHAR(30)	No		Allowed action

Table 14

Data Dictionary of `security.role_permission` — Permissions of Each Role

Field	Type	Null	Key	Description
role_permission_id	BIGINT IDENTITY	No	PK	Assignment identifier
role_id	SMALLINT	No	FK → role; UK con permission_id	Role
permission_id	SMALLINT	No	FK → permission	Granted permission

Table 15

Data Dictionary of `security.user_role` — Roles of Each User

Field	Type	Null	Key	Description
user_role_id	BIGINT IDENTITY	No	PK	Assignment identifier
user_id	BIGINT	No	FK → app_user; UK con role_id	User
role_id	SMALLINT	No	FK → role	Assigned role

Table 16

Data Dictionary of `security.user_session` — Open Sessions

Field	Type	Null	Key	Description
user_session_id	BIGINT IDENTITY	No	PK	Session identifier
user_id	BIGINT	No	FK → app_user	User
refresh_token_hash	NVARCHAR(255)	Yes		Refresh token hash
started_at	DATETIME2(3)	No		Session start (UTC)
expires_at	DATETIME2(3)	No		Expiry
revoked_at	DATETIME2(3)	Yes		Logout time
ip_address	NVARCHAR(45)	Yes		Source IP
user_agent	NVARCHAR(300)	Yes		Source browser or app

Table 17

Data Dictionary of `security.password_reset_token` — Password Recovery Codes

Field	Type	Null	Key	Description
password_reset_token_id	BIGINT IDENTITY	No	PK	Request identifier
user_id	BIGINT	No	FK → app_user	User who requested the code
token_hash	NVARCHAR(255)	No	UK	Hash of the 6-digit code
requested_at	DATETIME2(3)	No		Request date
expires_at	DATETIME2(3)	No		Expiry; CHECK `expires_at > requested_at`
used_at	DATETIME2(3)	Yes		When it was used
ip_address	NVARCHAR(45)	Yes		Request IP

Table 18

Data Dictionary of `security.user_preference` — UI and Notification Preferences

Field	Type	Null	Key	Description
user_preference_id	BIGINT IDENTITY	No	PK	Preference identifier
user_id	BIGINT	No	FK → app_user, UK	User (one row per user)
theme	NVARCHAR(20)	No		`green-light`, `green-dark`, `pink`, or `pink-dark`
language	NVARCHAR(5)	No		`es`, `en`, `fr`, or `pt`
notifications_enabled	BIT	No		In-app notifications
email_reminders_enabled	BIT	No		Email reminders
promotions_enabled	BIT	No		Promotion notices
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 19

Data Dictionary for `security.login_lockout` — Lockout After Failed Attempts

Field	Type	Null	Key	Description
user_id	BIGINT	No	PK, FK → app_user	Account with recent failures or locked
failed_count	INT	No		Failures within the current window, 0 to 4 (default 0)
first_failed_at	DATETIME2(3)	Yes		Start of the 15-minute window
locked_until	DATETIME2(3)	Yes		Locked until this time; NULL if not locked
updated_at	DATETIME2(3)	No		Last change

Table 20

Data Dictionary for `security.person_address` — Contact Address

Field	Type	Null	Key	Description
person_id	BIGINT	No	PK, FK → person	Person who owns the address
address	NVARCHAR(200)	No		Profile contact address; cannot be blank
updated_at	DATETIME2(3)	No		Last change

Schema `audit`

Change log. Owner: security-service.

Table 21

Data Dictionary of `audit.audit_log` — Change Log

Field	Type	Null	Key	Description
audit_log_id	BIGINT IDENTITY	No	PK	Record identifier
entity	NVARCHAR(60)	No		Affected entity
entity_id	BIGINT	No		Affected row
action	NVARCHAR(20)	No		Action performed
previous_data	NVARCHAR(MAX)	Yes		Previous state
new_data	NVARCHAR(MAX)	Yes		New state
changed_by	BIGINT	Yes		User who made the change
changed_at	DATETIME2(3)	No		Change date (UTC)

> *Note.* The table exists, but no trigger or service writes to it yet; who changed each row and when is kept by the audit block of each table (section 5.1).

Schema `customer`

Customers and vehicles. Owner: customer-service.

Table 22

Data Dictionary of `customer.vehicle_type` — Vehicle Types

Field	Type	Null	Key	Description
vehicle_type_id	SMALLINT IDENTITY	No	PK	Type identifier
code	NVARCHAR(30)	No	UK	`CAR`, `SEDAN`, `SUV`, `PICKUP`, `TRUCK`, or `MOTO`
name	NVARCHAR(60)	No		Display name
size_factor	DECIMAL(4,2)	No		Size factor (> 0)
display_order	SMALLINT	No		Display order
is_active	BIT	No		Type in use
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 23

Data Dictionary of `customer.customer` — Customers

Field	Type	Null	Key	Description
customer_id	BIGINT IDENTITY	No	PK	Customer identifier
person_id	BIGINT	No	UK, ⇢ security.person	Customer's person
user_id	BIGINT	Yes	UK filtrada, ⇢ security.app_user	Customer's account
loyalty_points	INT	No		Planned display copy of the points balance (≥ 0); no flow updates it, the balance is read from `payment.loyalty_transaction`
customer_since	DATE	No		Sign-up date
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 24

Data Dictionary of `customer.customer_vehicle` — Customers' Vehicles

Field	Type	Null	Key	Description
customer_vehicle_id	BIGINT IDENTITY	No	PK	Vehicle identifier
customer_id	BIGINT	No	FK → customer	Owner
license_plate	NVARCHAR(10)	No	UK filtrada (activos)	License plate
vehicle_type_id	SMALLINT	No	FK → vehicle_type	Vehicle type
brand	NVARCHAR(50)	Yes		Make
model	NVARCHAR(50)	Yes		Model
color	NVARCHAR(30)	Yes		Color
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Schema `catalog`

Service catalog and prices. Owner: booking-service.

Table 25

Data Dictionary of `catalog.service_category` — Service Categories

Field	Type	Null	Key	Description
service_category_id	SMALLINT IDENTITY	No	PK	Category identifier
code	NVARCHAR(30)	No	UK	`LAVADO`, `POLICHADO`, `DETALLADO`, `INTERIOR`, or `PAQUETES`
name	NVARCHAR(60)	No		Display name
description	NVARCHAR(200)	Yes		Description
display_order	SMALLINT	No		Display order
is_active	BIT	No		Category in use
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 26

Data Dictionary of `catalog.service` — Wash Services

Field	Type	Null	Key	Description
service_id	INT IDENTITY	No	PK	Service identifier
code	NVARCHAR(30)	No	UK	Code (`BASIC`, `PREMIUM`, `FULL` in the seed)
name	NVARCHAR(100)	No		Display name
description	NVARCHAR(MAX)	Yes		Description
service_category_id	SMALLINT	No	FK → service_category	Category
is_active	BIT	No		Available for booking
loyalty_points	INT	No		Points the service gives when the booking's payment is approved (changelog `022`)
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 27

Data Dictionary of `catalog.service_price` — Prices per Vehicle Type

Field	Type	Null	Key	Description
service_price_id	BIGINT IDENTITY	No	PK	Price identifier
service_id	INT	No	FK → service	Service
vehicle_type_id	SMALLINT	No	⇢ customer.vehicle_type	Vehicle type
price	DECIMAL(12,2)	No		Price (≥ 0)
estimated_minutes	SMALLINT	No		Estimated duration (> 0)
valid_from	DATE	No		Valid from
valid_to	DATE	Yes		Valid to; only one open price per service and type (filtered index); rows are immutable by trigger
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Schema `booking`

Establishment, business hours, bays, and bookings. Owner: booking-service.

Table 28

Data Dictionary of `booking.establishment` — Car Wash Data

Field	Type	Null	Key	Description
establishment_id	SMALLINT IDENTITY	No	PK	Identifier
legal_name	NVARCHAR(150)	No		Legal name
trade_name	NVARCHAR(150)	No		Trade name
tax_id	NVARCHAR(30)	No		Tax ID (NIT)
address	NVARCHAR(200)	No		Address
phone	NVARCHAR(20)	Yes		Phone

Field	Type	Null	Key	Description
email	NVARCHAR(120)	Yes		Email
logo_url	NVARCHAR(500)	Yes		Logo
business_type	NVARCHAR(100)	Yes		Business type (migration `032`)
founded_on	DATE	Yes		Foundation date; cannot be in the future (migration `032`)
legal_representative	NVARCHAR(150)	Yes		Legal representative (migration `032`)
legal_representative_document	NVARCHAR(30)	Yes		Legal representative ID (migration `032`)
website	NVARCHAR(200)	Yes		Website (migration `032`)
tax_regime	NVARCHAR(150)	Yes		Tax regime (migration `032`)
ciiu_activity	NVARCHAR(150)	Yes		CIIU economic activity (migration `032`)
dian_resolution	NVARCHAR(60)	Yes		DIAN invoicing resolution (migration `032`)
invoice_prefix	NVARCHAR(10)	Yes		Invoice prefix (migration `032`)
invoice_range	NVARCHAR(30)	Yes		Invoice range (migration `032`)
instagram	NVARCHAR(100)	Yes		Instagram (migration `032`)
facebook	NVARCHAR(150)	Yes		Facebook (migration `032`)
support_line	NVARCHAR(30)	Yes		Support line (migration `032`)
service_hours_note	NVARCHAR(150)	Yes		Service hours shown to the customer (migration `032`)
singleton	BIT	No	UK	Always 1: prevents a second row
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 29

Data Dictionary of `booking.business_hour` — Weekly Hours

Field	Type	Null	Key	Description
business_hour_id	SMALLINT IDENTITY	No	PK	Identifier
day_of_week	SMALLINT	No	UK	Day of week (1 to 7)
opens_at	TIME(0)	No		Opening time
closes_at	TIME(0)	No		Closing time (after opening)
is_active	BIT	No		Working day
break_starts_at	TIME(0)	Yes		Break start (changelog `018`)
break_ends_at	TIME(0)	Yes		Break end
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 30

Data Dictionary of `booking.business_hour_exception` — Schedule Exceptions

Field	Type	Null	Key	Description
business_hour_exception_id	INT IDENTITY	No	PK	Identifier
exception_date	DATE	No	UK	Special date
is_closed	BIT	No		Closed all day
opens_at	TIME(0)	Yes		Special opening (if open)
closes_at	TIME(0)	Yes		Special closing (if open)
reason	NVARCHAR(120)	No		Reason
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 31

Data Dictionary of `booking.bay_status` — Bay Statuses

Field	Type	Null	Key	Description
bay_status_id	SMALLINT IDENTITY	No	PK	Identifier
code	NVARCHAR(30)	No	UK	`ACTIVE`, `MAINTENANCE`, or `INACTIVE`
name	NVARCHAR(60)	No		Display name
accepts_bookings	BIT	No		Whether it accepts bookings
display_order	SMALLINT	No		Display order
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 32

Data Dictionary of `booking.service_bay` — Wash Bays

Field	Type	Null	Key	Description
service_bay_id	SMALLINT IDENTITY	No	PK	Bay identifier
code	NVARCHAR(20)	No	UK	Code (`BAY-01`…)
name	NVARCHAR(60)	No		Display name
bay_status_id	SMALLINT	No	FK → bay_status	Status (default 1, active)
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 33

Data Dictionary of `booking.booking_status` — Booking Statuses

Field	Type	Null	Key	Description
booking_status_id	SMALLINT IDENTITY	No	PK	1 `SCHEDULED`, 2 `CONFIRMED`, 3 `IN_PROGRESS`, 4 `COMPLETED`, 5 `CANCELLED`, 6 `NO_SHOW`
code	NVARCHAR(30)	No	UK	Status code
name	NVARCHAR(60)	No		Display name
is_final	BIT	No		Whether the status is final
display_order	SMALLINT	No		Display order
is_active	BIT	No		Status in use
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 34

Data Dictionary of `booking.cancellation_reason` — Cancellation Reasons

Field	Type	Null	Key	Description
cancellation_reason_id	SMALLINT IDENTITY	No	PK	Identifier
code	NVARCHAR(30)	No	UK	Code (eight seed reasons; `EXPIRED`, from seed `112`, is kept but no process uses it since `ExpiredBookingsJob` was removed on October 24, 2026 (B-07), and `PAYMENT_TIMEOUT` (*The payment receipt was not uploaded in time*), from seed `117` of October 26, 2026, is only used by the system when it cancels an overdue booking with digital payment; neither is offered when cancelling by hand)
name	NVARCHAR(60)	No		Display name
is_customer_fault	BIT	No		Whether the customer is at fault
display_order	SMALLINT	No		Display order
is_active	BIT	No		Reason in use
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 35

Data Dictionary of `booking.booking` — Bookings

Field	Type	Null	Key	Description
booking_id	BIGINT IDENTITY	No	PK	Booking identifier
customer_vehicle_id	BIGINT	No	⇢ customer.customer_vehicle	Vehicle to wash
service_bay_id	SMALLINT	Yes	FK → service_bay	Assigned bay; a trigger prevents overlapping times
scheduled_start	DATETIME2(3)	No		Scheduled start
scheduled_end	DATETIME2(3)	No		Scheduled end (after start)
booking_status_id	SMALLINT	No	FK → booking_status	Status
booked_by	BIGINT	No	⇢ security.app_user	User who created the booking
cancellation_reason_id	SMALLINT	Yes	FK → cancellation_reason	Only when the status is `CANCELLED`
points_redeemed	INT	No		Points redeemed
points_discount_amount	DECIMAL(12,2)	No		Points discount (both 0 or both > 0)
notes	NVARCHAR(300)	Yes		Notes

Field	Type	Null	Key	Description
payment_mode	NVARCHAR(10)	Yes		How the customer will pay: `DIGITAL` (QR code and receipt) or `CASH` (cash at the car wash), with CHECK `ck_booking_payment_mode`. NULL in bookings made before the change and in the administrator's bookings without a mode (migration `049`; NVARCHAR because of `use_nationalized_character_data`, the lesson of `048`)
payment_deadline	DATETIME2(3)	Yes		Deadline to upload the receipt of a `DIGITAL` booking: the earlier of creation plus `BOOKING_PAYMENT_WINDOW` (30 min) and `scheduled_start`. NULL if it has no deadline (migration `049`)
payment_reported_at	DATETIME2(3)	Yes		When the receipt or the approved payment arrived from payment-service (`payment.reported`, `payment.confirmed`, or the internal query). The filtered index `ix_booking_payment_pending` (`payment_deadline`, only `DIGITAL` without a report) serves `PaymentDeadlineJob` (migration `049`)
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 36

Data Dictionary of `booking.booking_service` — Service Lines of Each Booking

Field	Type	Null	Key	Description
booking_service_id	BIGINT IDENTITY	No	PK	Line identifier
booking_id	BIGINT	No	FK → booking; UK con service_price_id	Booking
service_price_id	BIGINT	No	FK → catalog.service_price	Price applied when booking
quantity	SMALLINT	No		Quantity (> 0)
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 37

Data Dictionary of `booking.schedule_history` — Schedule Change History

Field	Type	Null	Key	Description
schedule_history_id	BIGINT IDENTITY	No	PK	Identifier
entity_type	NVARCHAR(20)	No		`WEEK`, `EXCEPTION`, or `BAY`
title	NVARCHAR(200)	No		Change summary
detail	NVARCHAR(300)	Yes		Change detail
changed_at	DATETIME2(3)	No		Change date (UTC)
changed_by	BIGINT	Yes		User who made the change

Schema `execution`

Operators and service execution. Owner: operations-service.

Table 38

Data Dictionary of `execution.execution_status` — Execution Statuses

Field	Type	Null	Key	Description
execution_status_id	SMALLINT IDENTITY	No	PK	1 `PENDING`, 2 `IN_PROGRESS`, 3 `PAUSED`, 4 `COMPLETED`, 5 `WITH_ISSUE`
code	NVARCHAR(30)	No	UK	Status code
name	NVARCHAR(60)	No		Display name
is_final	BIT	No		Whether the status is final
display_order	SMALLINT	No		Display order
is_active	BIT	No		Status in use
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 39

Data Dictionary of `execution.operator` — Operators

Field	Type	Null	Key	Description
operator_id	INT IDENTITY	No	PK	Operator identifier
user_id	BIGINT	No	UK, ⇢ security.app_user	Operator's account
hired_on	DATE	No		Hire date
is_active	BIT	No		Eligible for assignments
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 40

Data Dictionary of `execution.operator_availability` — Operator Weekly Shift

Field	Type	Null	Key	Description
operator_availability_id	INT IDENTITY	No	PK	Identifier
operator_id	INT	No	FK → operator; UK con day_of_week	Operator
day_of_week	SMALLINT	No		Day of week (1 to 7)
starts_at	TIME(0)	No		Shift start
ends_at	TIME(0)	No		Shift end (after start)
is_active	BIT	No		Shift in use
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 41

Data Dictionary of `execution.operator_absence` — Operator Absences

Field	Type	Null	Key	Description
operator_absence_id	INT IDENTITY	No	PK	Identifier
operator_id	INT	No	FK → operator	Operator
starts_at	DATETIME2(3)	No		Absence start
ends_at	DATETIME2(3)	No		Absence end; a trigger prevents overlapping absences
reason	NVARCHAR(120)	No		Reason
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 42

Data Dictionary of `execution.service_execution` — Execution of Each Service Line

Field	Type	Null	Key	Description
service_execution_id	BIGINT IDENTITY	No	PK	Execution identifier
booking_service_id	BIGINT	No	UK, ⇢ booking.booking_service	Booking line
operator_id	INT	No	FK → operator	Assigned operator
execution_status_id	SMALLINT	No	FK → execution_status	Execution status
started_at	DATETIME2(3)	Yes		Actual start
finished_at	DATETIME2(3)	Yes		Actual finish
quality_rating	SMALLINT	Yes		Rating from 1 to 5
quality_comment	NVARCHAR(500)	Yes		Comment; only with a rating
rated_at	DATETIME2(3)	Yes		Rating date
is_comment_visible	BIT	No		Whether the comment is shown
follow_up_notice_at	DATETIME2(3)	Yes		When the operator and the administrators were told that the service ended without being registered; cleared when the service starts (changelog `040`)
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 43

Data Dictionary for `execution.operator_profile` — Operator Specialty

Field	Type	Null	Key	Description
operator_id	INT	No	PK, FK → operator	Operator
specialty	NVARCHAR(80)	Yes		Specialty
updated_at	DATETIME2(3)	No		Last change
updated_by	BIGINT	Yes		Administrator who changed it ⇢ security.app_user

Table 44

Data Dictionary for `execution.operator_skill` — Operator Skills

Field	Type	Null	Key	Description
operator_skill_id	INT IDENTITY	No	PK	Identifier
operator_id	INT	No	FK → operator	Operator
name	NVARCHAR(60)	No	UK with operator_id	Skill name
skill_level	NVARCHAR(12)	No		`CERTIFIED`, `EXPERT` or `ADVANCED`
display_order	SMALLINT	No		Order in the profile

Table 45

Data Dictionary for `execution.inspection_report` — Vehicle Inspection Report

Field	Type	Null	Key	Description
inspection_report_id	BIGINT IDENTITY	No	PK	Identifier
booking_id	BIGINT	No	UK, ⇢ booking.booking	Inspected booking (one report per booking)
booking_code, vehicle, plate, services	NVARCHAR	Yes		Copy of the booking data when the report is created
booking_date, booking_time, duration_minutes	DATE, TIME(0), INT	No		Booking date, time, and duration
public_token	NVARCHAR(64)	No	UK	Random code of the public link
is_published	BIT	No		The link shows the report (default 0)
published_at	DATETIME2(3)	Yes		Publication date; required when published
created_at, created_by, updated_at, updated_by, row_version	—	—		Audit

Table 46

Data Dictionary for `execution.inspection_phase` — Report Phases

Field	Type	Null	Key	Description
inspection_phase_id	BIGINT IDENTITY	No	PK	Identifier
inspection_report_id	BIGINT	No	FK → inspection_report; UK with phase	Report
phase	NVARCHAR(20)	No		`RECEPTION`, `EXTERIOR`, `INTERIOR`, `ENGINE` or `DELIVERY`
status	NVARCHAR(20)	No		`PENDING`, `IN_PROGRESS` or `DONE`
started_at, finished_at	DATETIME2(3)	Yes		Phase start and end

Table 47

Data Dictionary for `execution.inspection_finding` — Report Findings

Field	Type	Null	Key	Description
inspection_finding_id	BIGINT IDENTITY	No	PK	Identifier
inspection_report_id	BIGINT	No	FK → inspection_report	Report
phase	NVARCHAR(20)	No		Phase where it was found
area	NVARCHAR(120)	No		Vehicle area
severity	NVARCHAR(10)	No		`INFO`, `MINOR` or `MAJOR`
note	NVARCHAR(500)	Yes		Note
created_at, created_by, deleted_at, deleted_by	—	—		Creation and soft delete

Table 48

Data Dictionary for `execution.inspection_photo` — Finding Photos

Field	Type	Null	Key	Description
inspection_photo_id	BIGINT IDENTITY	No	PK	Identifier
inspection_finding_id	BIGINT	No	FK → inspection_finding	Finding (up to 5 photos each)
content_type	NVARCHAR(30)	No		`image/jpeg` or `image/png`
content	VARBINARY(MAX)	No		Image
size_bytes	INT	No		Size, up to 2 MB
created_at, created_by	—	—		Creation

Schema `payment`

Payments, payment accounts, and points. Owner: payment-service.

Table 49

Data Dictionary of `payment.payment_method_type` — Payment Methods

Field	Type	Null	Key	Description
payment_method_type_id	SMALLINT IDENTITY	No	PK	Identifier
code	NVARCHAR(30)	No	UK	`EFECTIVO`, `NEQUI`, `DAVIPLATA`, or `TRANSFERENCIA`
name	NVARCHAR(60)	No		Display name
requires_account	BIT	No		Whether it requires an account
requires_receipt	BIT	No		Whether it requires a receipt
display_order	SMALLINT	No		Display order
is_active	BIT	No		Method in use
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 50

Data Dictionary of `payment.payment_account` — Car Wash Receiving Accounts

Field	Type	Null	Key	Description
payment_account_id	SMALLINT IDENTITY	No	PK	Account identifier
payment_method_type_id	SMALLINT	No	FK → payment_method_type	Payment method
account_holder	NVARCHAR(120)	No		Holder
account_number	NVARCHAR(50)	Yes		Number or key
qr_image_url	NVARCHAR(MAX)	Yes		QR image stored as data (changelog `016`)
instructions	NVARCHAR(300)	Yes		Instructions for the customer
is_active	BIT	No		Account visible to the customer
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 51

Data Dictionary of `payment.payment_status` — Payment Statuses

Field	Type	Null	Key	Description
payment_status_id	SMALLINT IDENTITY	No	PK	Identifier
code	NVARCHAR(30)	No	UK	`PENDING`, `IN_REVIEW`, `APPROVED`, `REJECTED`, or `REFUNDED`
name	NVARCHAR(60)	No		Display name
is_final	BIT	No		Whether the status is final
display_order	SMALLINT	No		Display order
is_active	BIT	No		Status in use
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 52

Data Dictionary of `payment.payment` — Payments

Field	Type	Null	Key	Description
payment_id	BIGINT IDENTITY	No	PK	Payment identifier
booking_id	BIGINT	No	⇢ booking.booking	Booking paid; one approved payment per booking (filtered index)
payment_account_id	SMALLINT	No	FK → payment_account	Account paid into
payment_status_id	SMALLINT	No	FK → payment_status	Status
amount	DECIMAL(12,2)	No		Amount (> 0)
processed_at	DATETIME2(3)	Yes		Review date
approved_by	BIGINT	Yes		Approving administrator
rejection_reason	NVARCHAR(200)	Yes		Rejection reason
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 53

Data Dictionary of `payment.payment_receipt` — Payment Receipts

Field	Type	Null	Key	Description
payment_receipt_id	BIGINT IDENTITY	No	PK	Receipt identifier
payment_id	BIGINT	No	FK → payment	Payment
file_url	NVARCHAR(MAX)	No		Receipt image
transaction_reference	NVARCHAR(100)	Yes		Transaction reference
reported_amount	DECIMAL(12,2)	Yes		Amount declared by the customer (> 0)
uploaded_at	DATETIME2(3)	No		Upload date
uploaded_by	BIGINT	No		User who uploaded it
reviewed_at	DATETIME2(3)	Yes		Review date
reviewed_by	BIGINT	Yes		Reviewer (together with reviewed_at)
review_comment	NVARCHAR(300)	Yes		Review comment
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 54

Data Dictionary of `payment.loyalty_movement_type` — Points Movement Types

Field	Type	Null	Key	Description
loyalty_movement_type_id	SMALLINT IDENTITY	No	PK	Identifier
code	NVARCHAR(30)	No	UK	`EARNED`, `REDEEMED`, `EXPIRED`, `ADJUSTED`, or `REVERSED`
name	NVARCHAR(60)	No		Display name
sign	SMALLINT	No		+1 or -1
display_order	SMALLINT	No		Display order
is_active	BIT	No		Type in use
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 55

Data Dictionary of `payment.loyalty_transaction` — Points Ledger

Field	Type	Null	Key	Description
loyalty_transaction_id	BIGINT IDENTITY	No	PK	Movement identifier
customer_id	BIGINT	No	⇢ security.app_user; UK with sequence_no	Customer's user id (the name comes from the first migration, ADR-015)
loyalty_movement_type_id	SMALLINT	No	FK → loyalty_movement_type	Movement type
booking_id	BIGINT	Yes	⇢ booking.booking	Booking that caused the movement
points	INT	No		Movement points (≠ 0)
balance_after	INT	No		Resulting balance (≥ 0); source of truth for the balance
expires_on	DATE	Yes		Points expiry
description	NVARCHAR(200)	Yes		Detail
sequence_no	INT	No		Per-customer sequence number (changelog `020`)
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Schema `promotion`

Promotions and coupons. Owner: payment-service.

Table 56

Data Dictionary of `promotion.discount_type` — Discount Types

Field	Type	Null	Key	Description
discount_type_id	SMALLINT IDENTITY	No	PK	Identifier
code	NVARCHAR(30)	No	UK	`PERCENT`, `FIXED`, or `PACKAGE`
name	NVARCHAR(60)	No		Display name
display_order	SMALLINT	No		Display order
is_active	BIT	No		Type in use
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 57

Data Dictionary of `promotion.promotion` — Promotions

Field	Type	Null	Key	Description
promotion_id	INT IDENTITY	No	PK	Promotion identifier
code	NVARCHAR(30)	No	UK	Coupon code
name	NVARCHAR(100)	No		Display name
description	NVARCHAR(300)	Yes		Description
discount_type_id	SMALLINT	No	FK → discount_type	Discount type
discount_value	DECIMAL(12,2)	No		Discount value (> 0)

Field	Type	Null	Key	Description
max_discount_amount	DECIMAL(12,2)	Yes		Discount cap
min_purchase_amount	DECIMAL(12,2)	No		Minimum purchase
is_public	BIT	No		Visible to all customers (the code does not read it yet: every active promotion is offered)
min_completed_booking	INT	No		Completed bookings required (≥ 0)
valid_from	DATE	No		Valid from
valid_to	DATE	No		Valid to
max_redemptions	INT	Yes		Total redemption limit
max_redemptions_per_customer	INT	Yes		Per-customer limit (≤ total)
is_active	BIT	No		Promotion active
price	DECIMAL(12,2)	Yes		Package price (changelog `017`)
duration_minutes	INT	Yes		Package duration
icon	NVARCHAR(40)	Yes		Icon
featured	BIT	No		Featured
benefits	NVARCHAR(600)	Yes		Package benefits
required_points	INT	No		Points needed to redeem (≥ 0; changelog `018`)
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 58

Data Dictionary of `promotion.promotion_service` — Services Each Promotion Applies To

Field	Type	Null	Key	Description
promotion_service_id	INT IDENTITY	No	PK	Identifier
promotion_id	INT	No	FK → promotion; UK con service_id	Promotion
service_id	INT	No	⇢ catalog.service	Service
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 59

Data Dictionary of `promotion.promotion_customer` — Target Customers of Each Promotion

Field	Type	Null	Key	Description
promotion_customer_id	BIGINT IDENTITY	No	PK	Identifier
promotion_id	INT	No	FK → promotion; UK con customer_id	Promotion
customer_id	BIGINT	No	⇢ customer.customer	Target customer
assigned_at	DATETIME2(3)	No		Assignment date
assigned_by	BIGINT	Yes		Assigning user
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 60

Data Dictionary of `promotion.booking_promotion` — Promotion Redemptions on Bookings

Field	Type	Null	Key	Description
booking_promotion_id	BIGINT IDENTITY	No	PK	Redemption identifier
booking_id	BIGINT	No	⇢ booking.booking; UK con promotion_id	Booking
promotion_id	INT	No	FK → promotion	Redeemed promotion
applied_amount	DECIMAL(12,2)	No		Discount applied (> 0)
discount_percent	DECIMAL(5,2)	Yes		Percentage of a percent coupon (> 0 and ≤ 100) used to recalculate the discount with the current booking total (BIZ-05, changelog `022`); NULL for fixed-amount coupons
max_discount_amount	DECIMAL(12,2)	Yes		Cap in pesos of a percent coupon, if the promotion had one (> 0, changelog `022`)
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Schema `notification`

Notifications, devices, reminders, and pending push and email deliveries. Owner: notification-service.

Table 61

Data Dictionary of `notification.notification_type` — Notification Types

Field	Type	Null	Key	Description
notification_type_id	SMALLINT IDENTITY	No	PK	Identifier
code	NVARCHAR(40)	No	UK	One of the 26 codes, from `USER_WELCOME` to `ADMIN_SERVICE_NOT_REGISTERED` (changelogs `038` and `041`)
name	NVARCHAR(80)	No		Display name
display_order	SMALLINT	No		Display order
is_active	BIT	No		Type in use
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 62

Data Dictionary of `notification.notification` — Notifications

Field	Type	Null	Key	Description
notification_id	BIGINT IDENTITY	No	PK	Notification identifier
user_id	BIGINT	No	⇢ security.app_user	Recipient
notification_type_id	SMALLINT	No	FK → notification_type	Type
title	NVARCHAR(120)	No		Title
message	NVARCHAR(500)	No		Message
reference_entity	NVARCHAR(40)	Yes		Related entity (for example, booking)
reference_id	BIGINT	Yes		Related entity identifier
is_read	BIT	No		Read (kept consistent with read_at by a CHECK)
read_at	DATETIME2(3)	Yes		Read date
sent_at	DATETIME2(3)	No		Send date
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 63

Data Dictionary of `notification.device_token` — Push Notification Devices

Field	Type	Null	Key	Description
device_token_id	BIGINT IDENTITY	No	PK	Device identifier
user_id	BIGINT	No	⇢ security.app_user	Owner
token	NVARCHAR(200)	No	UK	Expo Push token
platform	NVARCHAR(10)	No		Platform; the CHECK accepts `ANDROID` or `IOS`
is_active	BIT	No		Device in use
last_seen_at	DATETIME2(3)	No		Last device registration
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 64

Data Dictionary of `notification.processed_event` — Already Handled Events

Field	Type	Null	Key	Description
processed_event_id	BIGINT IDENTITY	No	PK	Identifier
event_id	NVARCHAR(64)	No	UK	RabbitMQ event ID
event_type	NVARCHAR(60)	No		Event type
processed_at	DATETIME2(3)	No		Processing date
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 65

Data Dictionary of `notification.booking_reminder` — Booking Reminders

Field	Type	Null	Key	Description
booking_reminder_id	BIGINT IDENTITY	No	PK	Reminder identifier
booking_id	BIGINT	No	⇢ booking.booking; UK con lead_minutes	Booking
user_id	BIGINT	No	⇢ security.app_user	Customer
booking_code	NVARCHAR(20)	Yes		Booking code
scheduled_start	DATETIME2(3)	No		Appointment time
lead_minutes	INT	No		Minutes of notice (`REMINDER_LEADS`, default 24 h and 1 h)
remind_at	DATETIME2(3)	No		When to notify
sent_at	DATETIME2(3)	Yes		When it was sent
created_at, created_by, updated_at, updated_by, deleted_at, deleted_by, row_version	—	—		Audit block (section 5.1)

Table 66

Data Dictionary of `notification.booking_tracking` — Local Copy of Booking State

Field	Type	Null	Key	Description
booking_id	BIGINT	No	PK, ⇢ booking.booking	Booking
booking_code	NVARCHAR(20)	Yes		Booking code
customer_user_id	BIGINT	Yes		Customer user
operator_user_id	BIGINT	Yes		Assigned operator user
scheduled_start	DATETIME2(3)	Yes		Appointment time
is_cancelled	BIT	No		Booking cancelled
updated_at	DATETIME2(3)	No		Last update
unassigned_alert_at	DATETIME2(3)	Yes		When the "no operator" alert was sent (changelog `031`)

Table 67

Data Dictionary of `notification.pending_delivery` — Push and Email Deliveries Pending or to Retry

Field	Type	Null	Key	Description
pending_delivery_id	BIGINT IDENTITY	No	PK	Delivery identifier
notification_id	BIGINT	No	FK → notification; UK with channel	Notification being delivered
channel	NVARCHAR(10)	No	UK with notification_id	`PUSH` or `EMAIL`; a single delivery per notification and channel (NVARCHAR since changelog `048`)
email_address	NVARCHAR(254)	Yes		Destination email (`EMAIL` channel)
recipient_name	NVARCHAR(100)	Yes		Recipient name
language	NVARCHAR(5)	Yes		Message language (NVARCHAR since changelog `048`)
attempts	INT	No		Attempts made (≥ 0, default 0)
next_attempt_at	DATETIME2(3)	No		When it is tried again; filtered index of the ones still pending
last_error	NVARCHAR(300)	Yes		Last error
delivered_at	DATETIME2(3)	Yes		When it was delivered
failed_at	DATETIME2(3)	Yes		When it was given up after reaching the retry cap
created_at	DATETIME2(3)	No		Creation (default `SYSUTCDATETIME()`)

The notification and its deliveries are written in the same transaction and go out after the commit; if they fail, a scheduled job retries them up to a cap (changelog `047`, OPS-05).

> *Note.* Source of types and constraints: `.sql` files in `*/src/main/resources/db/changelog/` (Java services) and `lavarapido-payment-service/db/changelog/changes/`, including the `ALTER` statements made after each table was created.

6. Class Diagrams

There is at least one class diagram for each of the six services; the four largest have two: one for the domain model and one for the application layer with its ports. Business rules live in the domain entities, not in the controllers: for example, `Booking.moveTo()` decides which status changes are valid, and `ServiceExecution.rate()` allows rating only once and only for a completed service.

Diagram conventions:

- UML stereotypes: `aggregate root`, `entity`, `value object`, `enumeration`, `domain service`, `application service`, `interface` (outbound port), `event`, `factory`, and `abstract`.
- The `implements …` line in an application service lists the inbound ports (use cases) it implements.
- A solid arrow (`-->`) is a constructor-injected dependency; a dashed arrow (`..>`) is a usage or a reference by ID.
- Getters, constructors, exceptions, response DTOs (`*View`, `*Dto`), and input commands (`*Command`, `*Request`) are omitted.

6.1 security-service

Figure 20

security-service Domain Model

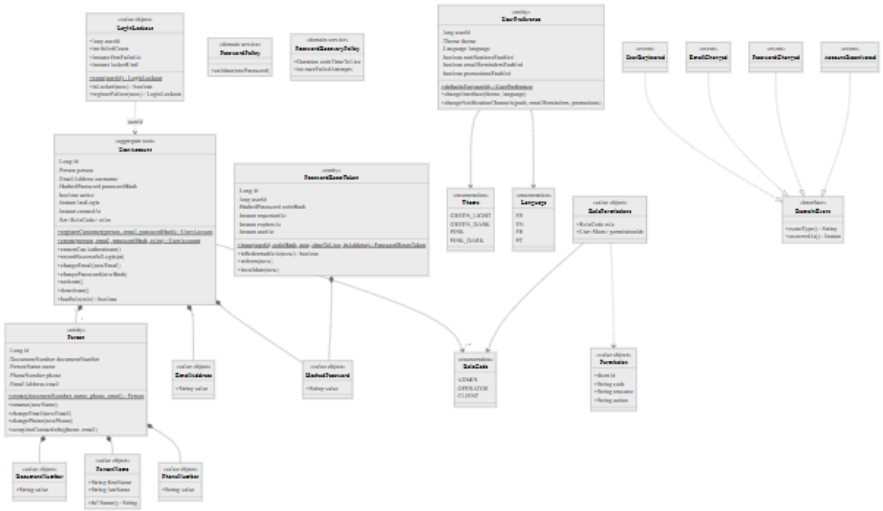
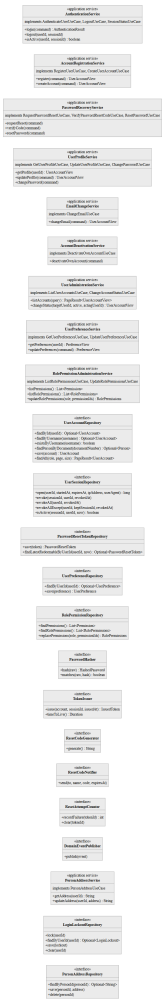


Figure 21

security-service Application Services and Ports



Note. Outbound adapters: `UserAccountPersistenceAdapter`, `UserSessionPersistenceAdapter`, `PasswordResetTokenPersistenceAdapter`, `UserPreferencePersistenceAdapter`, `RolePermissionPersistenceAdapter`, `LoginLockoutPersistenceAdapter`, and `PersonAddressPersistenceAdapter` (JPA); `BCryptPasswordHasher`; `JwtTokenIssuer`; `SecureRandomResetCodeGenerator`; `EmailResetCodeNotifier`, `LoggingResetCodeNotifier`, and `UndeliveredResetCodeNotifier`; `InMemoryResetAttemptCounter`; `RabbitDomainEventPublisher` and `LoggingDomainEventPublisher`. Inbound adapters: `AuthController` (includes `GET /auth/session` for the gateway), `ActiveSessionValidator` (rejects the JWT of a closed session), `UserProfileController`, `UserAdministrationController`, `RolePermissionController`, `UserAddressController`, and `InternalUserController`. Source: `security-service/src/main/java/com/lavarapido/security/`.

Ports used by each service.

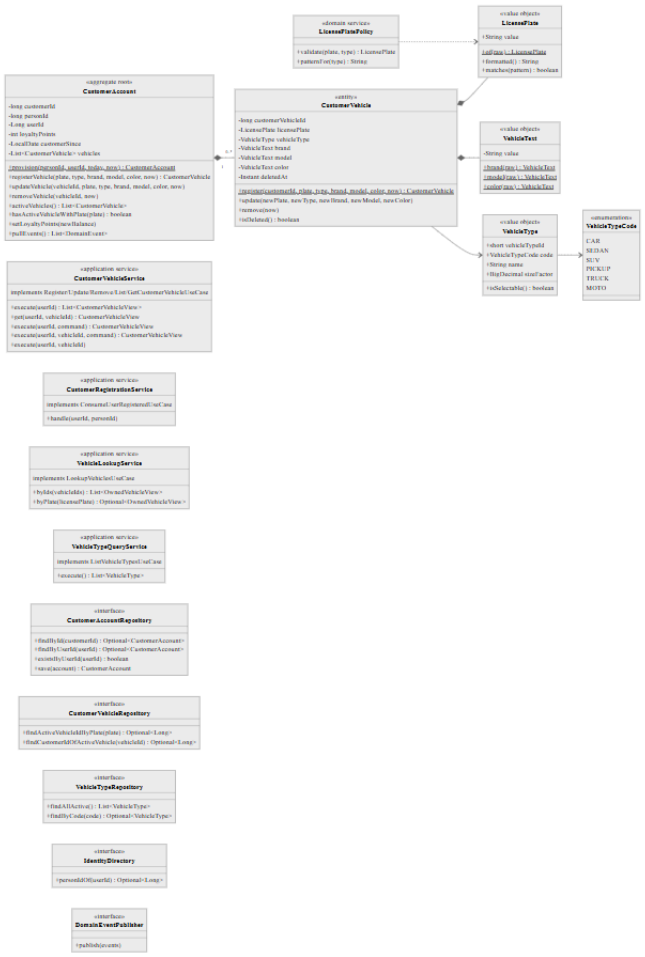
- `AuthenticationService`: `UserAccountRepository`, `LoginLockoutRepository`, `UserSessionRepository`, `PasswordHasher`, `TokenIssuer`.
- `AccountRegistrationService`: `UserAccountRepository`, `PasswordHasher`, `DomainEventPublisher`.

- `PasswordRecoveryService`: `LoginLockoutRepository`, `UserAccountRepository`, `PasswordResetTokenRepository`, `ResetCodeGenerator`, `ResetCodeNotifier`, `ResetAttemptCounter`, `PasswordHasher`.
- `UserProfileService`: `UserAccountRepository`, `PasswordHasher`.
- `EmailChangeService`: `UserAccountRepository`, `DomainEventPublisher`.
- `AccountDeactivationService`: `UserAccountRepository`, `UserSessionRepository`, `PasswordHasher`, `DomainEventPublisher`.
- `UserAdministrationService`: `UserAccountRepository`, `UserSessionRepository`, `DomainEventPublisher`.
- `UserPreferenceService`: `UserPreferenceRepository`.
- `RolePermissionAdministrationService`: `RolePermissionRepository`.
- `PersonAddressService`: `UserAccountRepository`, `PersonAddressRepository`.

6.2 customer-service

Figure 22

customer-service Classes



> *Note.* `Register/Update/Remove/List/GetCustomerVehicleUseCase` abbreviates the five vehicle inbound ports. Outbound adapters: `CustomerAccountPersistenceAdapter`, `CustomerVehiclePersistenceAdapter`, and `VehicleTypePersistenceAdapter` (JPA); `SecurityServiceIdentityDirectory` (REST to security-service); `LoggingDomainEventPublisher`, which only writes events to the log. Inbound adapters: `CustomerVehicleController`, `AdminVehicleController`, `VehicleTypeController`, and `UserRegisteredListener` (RabbitMQ). Source: `lavarapido-customer-service/src/main/java/com/lavarapido/customer/`.

What each application service uses.

- `CustomerVehicleService`: `CustomerAccount`, `CustomerAccountRepository`, `CustomerVehicleRepository`, `VehicleTypeRepository`, `IdentityDirectory`, `LicensePlatePolicy`, `DomainEventPublisher`.
- `CustomerRegistrationService`: `CustomerAccountRepository`, `DomainEventPublisher`.
- `VehicleLookupService`: `CustomerAccountRepository`, `CustomerVehicleRepository`. Since October 27, 2026 (*v1.1.1*), `byIds` gets the owners of all the vehicles in a single query (`findCustomerIdsOfActiveVehicles`) and loads each account once, instead of one query per vehicle.
- `VehicleTypeQueryService`: `VehicleTypeRepository`.

6.3 booking-service

Figure 23

booking-service Domain Model

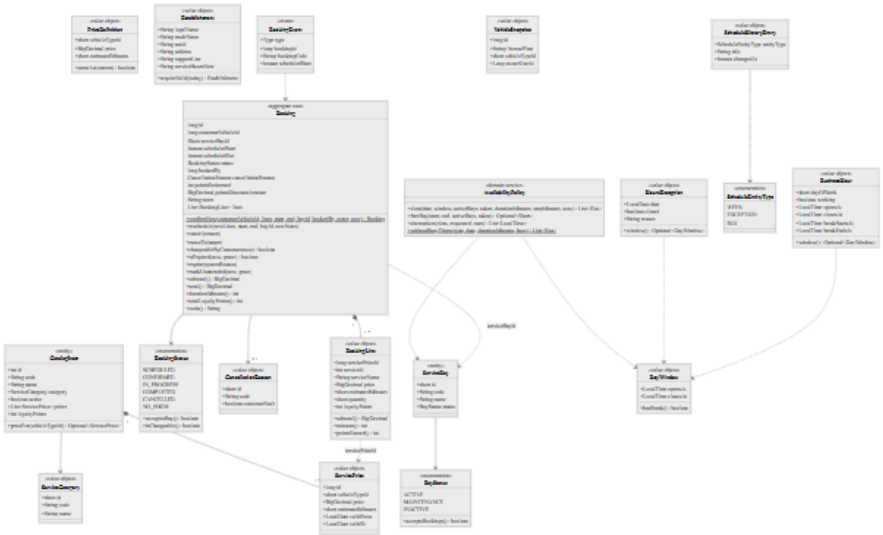


Figure 24

booking-service Application Services and Ports



Note. Outbound adapters: `BookingPersistenceAdapter`, `CatalogPersistenceAdapter`, `SchedulePersistenceAdapter`, and `BayPersistenceAdapter` (JPA); `CustomerServiceDirectory` (REST to customer-service); `RabbitDomainEventPublisher` and `LoggingDomainEventPublisher`. Inbound adapters: `BookingController`, `AdminBookingController`, `OperatorBookingController`, `CatalogController`, `ScheduleController`, and `InternalBookingController` (`/internal/v1/bookings`, with `X-Internal-Key`, including `POST /internal/v1/bookings/{id}/no-show`, which calls `markUnattended`); `PaymentEventsListener` (queue `booking.payment-events`) and `PaymentDeadlineJob` (scheduled job that calls `paymentOverdueBookings` and `cancelForPaymentTimeout`), since October 26, 2026; `PaymentServiceDirectory` (REST to payment-service, with `X-Internal-Key`) implements the outbound port `PaymentStatusDirectory`. `ExpiredBookingsJob` was removed on October 24, 2026 (B-07). Since October 27, 2026 (*v1.1.1*), `BookingDtos` accepts at most 10 `serviceIds` with no nulls (`MAX_SERVICES`, when creating and rescheduling) and returns 400 `VALIDATION_ERROR` instead of 500, as it does with an invalid day in the opening-hours week; `BookingResponse.from(view, includeOwner)` only fills `bookedBy` and the vehicle owner in the staff views (administrator and operator), not in the customer's; and the alternative times go through `AvailabilityPolicy.withoutBusyTimes`, which removes those that clash with another booking of the same vehicle. Source: `lavarapido-booking-service/src/main/java/com/lavarapido/booking/`.

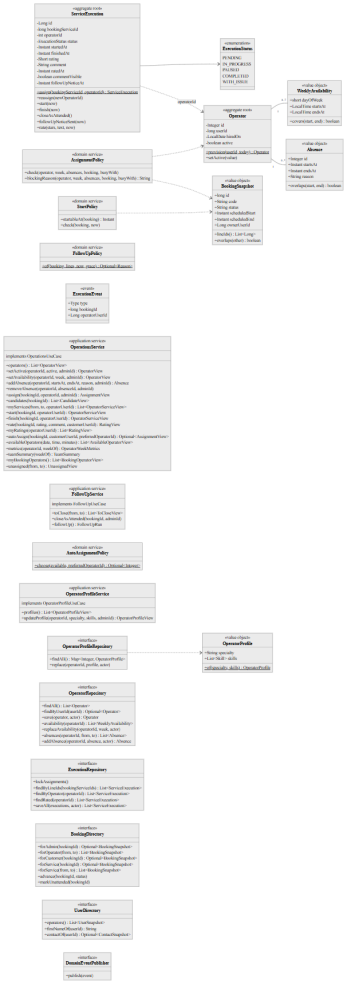
Ports used by each service.

- `BookingService`: `BookingRepository`, `CatalogRepository`, `ScheduleRepository`, `BayRepository`, `CustomerDirectory`, `DomainEventPublisher`, `BookingSettings`, and, since October 26, 2026, `PaymentStatusDirectory`. It also uses the domain policies `AvailabilityPolicy` and `PaymentPolicy` (cash limit and digital payment deadline; the mode is the `PaymentMode` value).
- `CatalogService`: `CatalogRepository`, `BookingSettings`.
- `ScheduleService`: `ScheduleRepository`, `BayRepository`, `BookingRepository`.

6.4 operations-service

Figure 25

operations-service Classes



Note. Outbound adapters: `OperatorPersistenceAdapter`, `ExecutionPersistenceAdapter`, `OperatorProfilePersistenceAdapter`, and `InspectionPersistenceAdapter` (JPA); `BookingServiceDirectory` (REST to booking-service, with the user's token or the internal key); `SecurityUserDirectory` (REST to security-service); `RabbitDomainEventPublisher` and `LoggingDomainEventPublisher`. Inbound adapters: `OperationsControllers` (administrator, operator, and rating routes), `AvailableOperatorsController`, `OperatorProfileController`, `InspectionControllers`, `FollowUpController` (`/admin/assignments/to-close` and

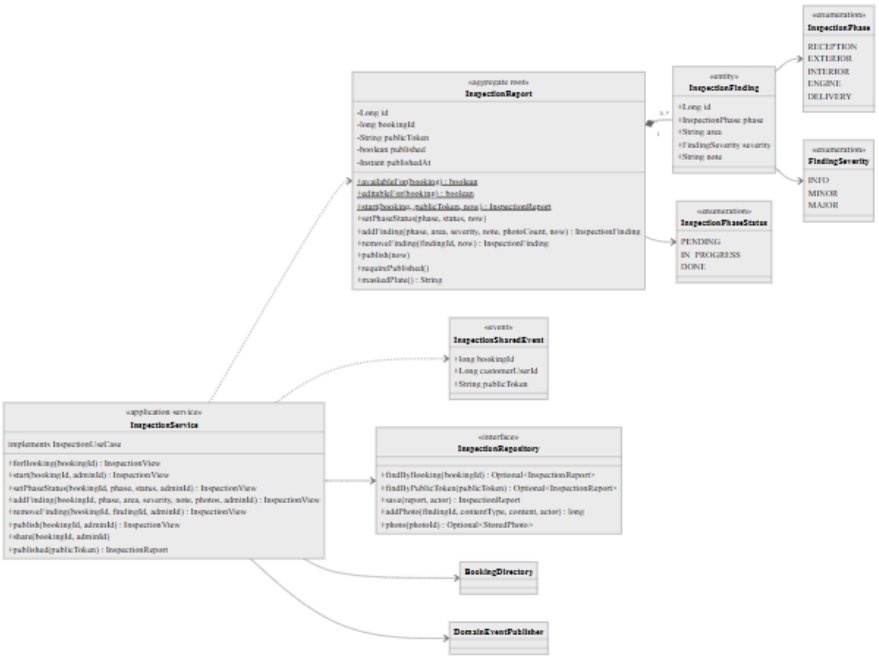
> `/admin/assignments/{bookingId}/attended`), `BookingConfirmedListener` (RabbitMQ, automatic assignment), and `FollowUpJob` (scheduled job that calls `followUp`). Source: `lavarapido-operations-service/src/main/java/com/lavarapido/operations/`.

What each application service uses.

- `OperationsService`: `OperatorRepository`, `ExecutionRepository`, `BookingDirectory`, `UserDirectory`, `DomainEventPublisher`, `AssignmentPolicy`, `StartPolicy`, `AutoAssignmentPolicy`, `ExecutionEvent`.
- `OperatorProfileService`: `OperatorProfileRepository`, `OperatorRepository`.
- `FollowUpService`: `ExecutionRepository`, `BookingDirectory`, `OperatorRepository`, `DomainEventPublisher`, `FollowUpPolicy`, `ExecutionEvent`.

Figure 26

Inspection Report in operations-service



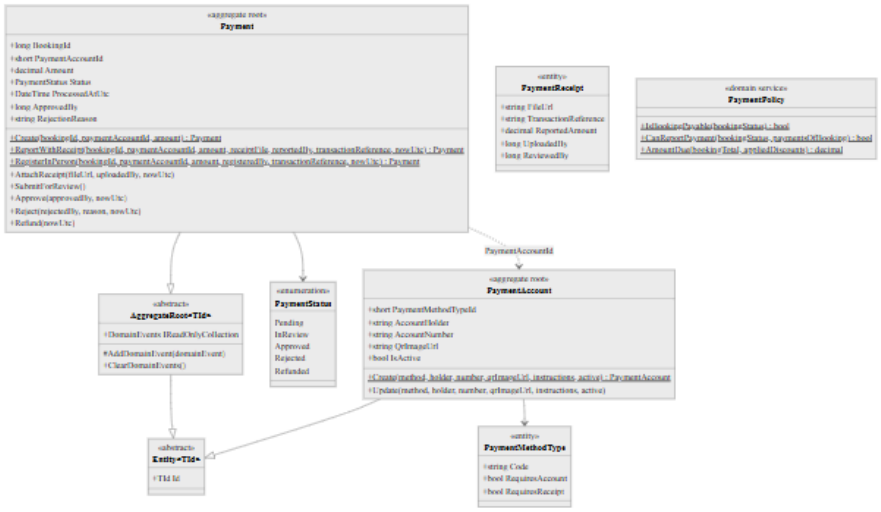
> *Note.* The report only exists for bookings of 180 minutes or more, can be edited while the booking is Confirmed or In progress, and can only be shared once published; the public link uses `publicToken` and shows a masked license plate. Adapters: `InspectionPersistenceAdapter` (JPA) and `InspectionControllers` (`/admin/inspections/**` and `/public/inspections/**` routes).

6.5 payment-service (.NET)

Figure 27

payment-service Domain Model

a) Payments



b) Promotions and loyalty (Promotion also inherits from `AggregateRoot<TId>`)

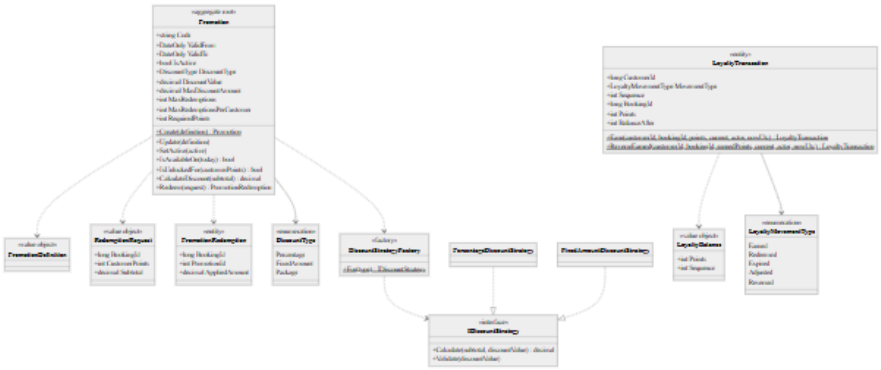
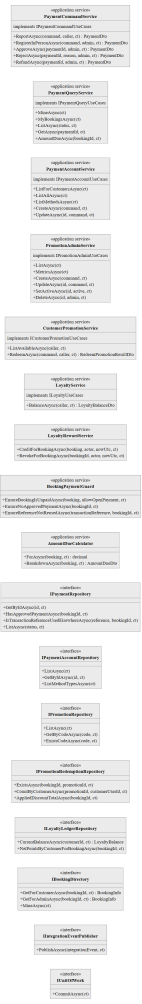


Figure 28

payment-service Application Services and Ports



Note. In .NET, attributes are public read-only properties and application methods are asynchronous (`Async`) and take a `CancellationToken ct`. Outbound adapters: `PaymentRepository`, `PaymentAccountRepository`, `PromotionRepository`, `PromotionRedemptionRepository`, `LoyaltyLedgerRepository`, and `EfUnitOfWork` (Entity Framework Core); `BookingServiceDirectory` (REST to booking-service); `RabbitMqIntegrationEventPublisher` and `NullIntegrationEventPublisher`. Inbound adapters: `PaymentsController`, `AdminPaymentsController`, `AdminPaymentAccountsController`, `PromotionsController`, and `LoyaltyController`; since October 26, 2026, `InternalPaymentsController` (`GET /internal/v1/payments/bookings/{bookingId}/reported`), protected by `InternalApiKeyAuthenticationHandler` (`X-Internal-Key` = `INTERNAL_API_KEY`). `Payment.ReportWithReceipt` raises the domain event `PaymentReported`, which goes out as `payment.reported`. Source: `lavarapido-payment-service/src/PaymentService.Domain` and `PaymentService.Application`.

Ports used by each service.

- `PaymentCommandService`: `IPaymentRepository`, `IBookingDirectory`, `IIntegrationEventPublisher`, `IUnitOfWork`, `BookingPaymentGuard`, `AmountDueCalculator`, `LoyaltyRewardService`.
- `PaymentQueryService`: `IPaymentRepository`, `IBookingDirectory`, `AmountDueCalculator`.

- PaymentAccountService: IPaymentAccountRepository, IUnitOfWork.
- PromotionAdminService: IPromotionRepository, IPromotionRedemptionRepository, IUnitOfWork.
- CustomerPromotionService: IPromotionRepository, IPromotionRedemptionRepository, ILoyaltyLedgerRepository, IBookingDirectory, BookingPaymentGuard, AmountDueCalculator, IUnitOfWork, IIntegrationEventPublisher.
- LoyaltyService: ILoyaltyLedgerRepository.
- LoyaltyRewardService: ILoyaltyLedgerRepository, IPromotionRepository.
- BookingPaymentGuard: IPaymentRepository, IPaymentAccountRepository.
- AmountDueCalculator: IPromotionRedemptionRepository.

6.6 notification-service

Figure 29

notification-service Domain Model

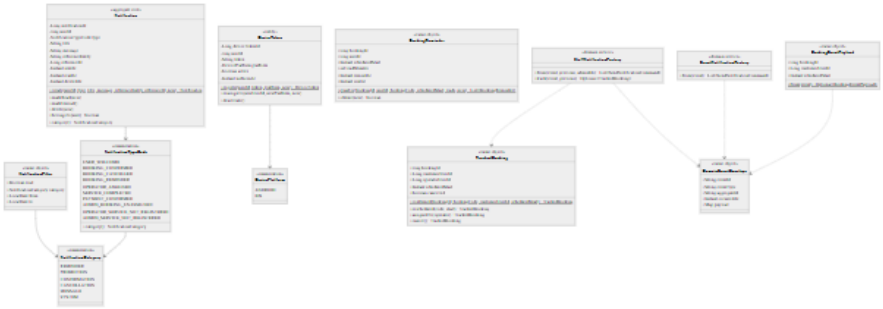
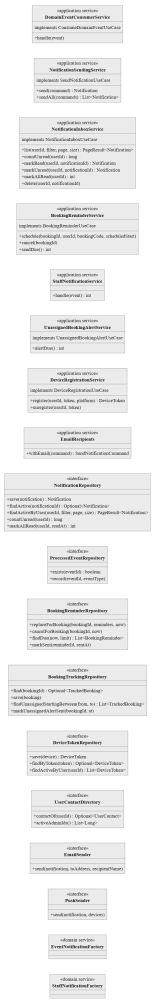


Figure 30

notification-service Application Services and Ports



Note. Between application services the real dependency is the inbound port: for example, `DomainEventConsumerService` receives a `SendNotificationUseCase`, implemented by `NotificationSendingService`. `NotificationTypeCode` has 26 codes; the diagram shows only some. Outbound adapters: `NotificationPersistenceAdapter`, `DeviceTokenPersistenceAdapter`, `BookingReminderPersistenceAdapter`, `BookingTrackingPersistenceAdapter`, and `ProcessedEventPersistenceAdapter` (JPA); `SmtpEmailSender` and `LoggingEmailSender`; `ExpoPushSender` and `LoggingPushSender`; `SecurityServiceUserContactDirectory` (internal REST to security-service). Inbound adapters: `NotificationController`, `AdminNotificationController`, `DeviceController`, `DomainEventListener` (RabbitMQ), and `ReminderScheduler` (scheduled job). Source: `lavarapido-notification-service/src/main/java/com/lavarapido/notification/`.

What each application service uses.

- `DomainEventConsumerService`: `ProcessedEventRepository`, `NotificationSendingService`, `EventNotificationFactory`, `BookingReminderService`, `StaffNotificationService`, `EmailRecipients`.
- `NotificationSendingService`: `NotificationRepository`, `DeviceTokenRepository`, `PushSender`, `EmailSender`, `UserContactDirectory`.
- `NotificationInboxService`: `NotificationRepository`.

- `BookingReminderService`: `BookingReminderRepository`, `NotificationSendingService`.
- `StaffNotificationService`: `BookingTrackingRepository`, `UserContactDirectory`, `StaffNotificationFactory`, `NotificationSendingService`.
- `UnassignedBookingAlertService`: `BookingTrackingRepository`, `UserContactDirectory`, `NotificationSendingService`.
- `DeviceRegistrationService`: `DeviceTokenRepository`.
- `EmailRecipients`: `UserContactDirectory`.

6.7 States of the Main Entities

Table 68

State Transitions Allowed by the Domain

Entity	Transition	Condition (code)
Booking	SCHEDULED → CONFIRMED	Has a bay assigned (`Booking.java:111`)
Booking	CONFIRMED → IN_PROGRESS or NO_SHOW	`Booking.java:112`
Booking	IN_PROGRESS → COMPLETED	`Booking.java:113`
Booking	→ CANCELLED	Only through `cancel(reason)`, not `moveTo` (`Booking.java:96, 114`)
Booking	Concurrent change	Since October 24, 2026 the booking has optimistic locking (`@Version` on `row_version`, changelog `045`): an old version answers 409 `CONFLICT`. A customer who reschedules only changes the date and time (409 `BOOKING_SERVICES_LOCKED`)
Booking	Ended without an operator: CONFIRMED → NO_SHOW	One hour after its end time and only if no line ever had an operator; operations asks for it and booking checks the expiry again (`Booking.java:141-147`, `FollowUpPolicy.java`)
Booking	Overdue digital payment: CONFIRMED → CANCELLED	`DIGITAL` booking without `payment_reported_at` whose `payment_deadline` has passed, and payment-service answers that it has no payment in review or approved; reason `PAYMENT_TIMEOUT` (`Booking.paymentOverdue`, `PaymentDeadlineJob.java`, since October 26, 2026)
Booking	Confirmed with an operator, not started or not finished	Never changes on its own: it goes to "To close" and the administrator resolves it (`FollowUpPolicy.java`, `FollowUpService.closeAsAttended`)
Execution	PENDING → IN_PROGRESS → COMPLETED	`start()` and `finish()` (`ServiceExecution.java:75-91`)
Execution	PENDING or IN_PROGRESS → COMPLETED ("Attended")	Only from "To close"; no made-up times (`ServiceExecution.java:98-103`)
Execution	Rating	Only when COMPLETED and only once (`ServiceExecution.java:111-129`)
Payment	Created → in review → approved, rejected, or refunded	Methods `SubmitForReview`, `Approve`, `Reject`, `Refund` (`Payment.cs:121-162`)
Execution	Start	From 15 minutes before the booking time and up to 4 hours after its end time (`START_LATE_LIMIT`, since October 24, 2026; 409 `SERVICE_TOO_LATE`) (`StartPolicy.java`)
Payment	Approval	Since October 24, 2026, if the amount does not match what is owed today, 409 `AMOUNT_CHANGED`; a single open payment per booking and a unique reference among live payments (migration 023)
Account	Locked for 15 minutes	On the fifth failure within 15 minutes; a correct sign-in or a password reset clears it (`LoginLockout.java`)
Inspection report	Draft → published	`publish()`; only a published report can be shared (`InspectionReport.java`)

Entity	Transition	Condition (code)
Inspection phase	PENDING ↔ IN_PROGRESS ↔ DONE	`setPhaseStatus()`; a finding in a pending phase moves it to IN_PROGRESS (`InspectionReport.java`)

Table 69

Inconsistencies Found When Comparing the Code With Earlier Documentation

#	Finding	Evidence
1	Technical Manual v2.0 says all six microservices are in Java 21; payment-service is in .NET 8	`lavarapido-payment-service/Directory.Build.props` (`net8.0`)
2	`MODELO-DE-DATOS.md` documented 43 tables and SRS v4 (Table 6) said 47; the changelogs created 57 as of October 15, 2026 and create 58 since notification changelog `047` (October 24, 2026)	Table 9 of this document
3	notification subscribed to `booking.created`, but booking does not publish that key. Fixed on October 24, 2026: the binding was removed (notification `8fde2ee`)	notification's `ROUTING_KEYS`
4	customer defines the `customer.provisioned`, `VehicleRegistered`, `VehicleUpdated`, `VehicleRemoved`, and `LoyaltyPointsChanged` events, but its only publisher writes them to the log: they never reach RabbitMQ and have no consumer	`CustomerProvisioned.java:15`, `customer/.../adapter/out/messaging/LoggingDomainEventPublisher.java`
5	Vehicle inspection, its public report, and the business data in Settings are simulated: they are stored in the browser, not in the backend	`Web/src/app/core/services/inspection-store.ts:20`, `features/admin/services/business-store.ts:52`; on mobile `scheduleMock.ts`, `settingsMock.ts`
6	In `docker-compose.yml`, the security and operations builds point to `../lavarapido-security-service` and `../lavarapido-operation-service`, which are the real repository names. In this local copy the folders are named `security-service` and `lavarapido-operations-service`, so `--profile app` fails unless they are cloned with their original names.	`Backend/lavarapido-infra/docker-compose.yml` (security-service and operations-service)
7	booking-service contains a `bin/src/main/java` folder in addition to `src/main/java` (an IDE-generated copy)	`Backend/lavarapido-booking-service/bin`
8	The cash account is not returned by `GET /payment-accounts` (only accounts that require a receipt); the web and mobile clients add it themselves. It is intentional, but the contract does not document it	`PaymentAccountService.cs:24-25`, `Web/.../client/pages/payment/payment.ts:44-57`
9	Hard-coded test text in the UI: "¡Hola, Laura!" on the administrator dashboard and "Laura Méndez" in payment review	`admin/pages/dashboard/dashboard.ts:73`, `payment-review-modal.html:167`
10	In the operator's "Assigned services" (web), the Customer column shows the license plate	`operator/pages/assigned-services/assigned-services.ts:141`

#	Finding	Evidence
11	The brand name is inconsistent across clients ("AquaWash", "Car Wash", "Lavado Vehicular", "Express Car Wash") and the landing page advertises a "mobile service… wherever you are", although SRS v4 defines a single location	Figures 2 and 7

Note. Code review of October 15, 2026: finding 5 is fixed (the inspection report, its public link, and the business data are stored in operations-service and booking-service; on mobile, `settingsMock.ts` only keeps the initial notification preferences) and so is finding 7 (`bin/` is now in booking-service's `.gitignore`). Findings 3, 4, 8, 9, 10, and 11 are still in the code. Review of October 20, 2026: finding 6 no longer applies (the local folders have the real repository names), finding 9 is no longer in the code, and finding 10 is fixed (the Customer column shows `clientName`); findings 3, 4, and 8 are unchanged. Review of October 21, 2026: finding 8 is fixed (`GET /payment-accounts` also returns the cash account while it is active, `PaymentAccountService.cs:27`, and the web and mobile apps show Cash only if it comes in that list); findings 3 and 4 are unchanged.

References

Angular. (n.d.). *Angular documentation*. Google. Retrieved October 9, 2026, from https://angular.dev/

Brown, S. (n.d.). *The C4 model for visualising software architecture*. Retrieved October 9, 2026, from https://c4model.com/

Cockburn, A. (2005). *Hexagonal architecture*. https://alistair.cockburn.us/hexagonal-architecture/

Evans, E. (2004). *Domain-driven design: Tackling complexity in the heart of software*. Addison-Wesley.

Expo. (n.d.). *Expo documentation*. Retrieved October 9, 2026, from https://docs.expo.dev/

Fowler, M. (2002). *Patterns of enterprise application architecture*. Addison-Wesley.

Gamma, E., Helm, R., Johnson, R., & Vlissides, J. (1994). *Design patterns: Elements of reusable object-oriented software*. Addison-Wesley.

Hohpe, G., & Woolf, B. (2003). *Enterprise integration patterns: Designing, building, and deploying messaging solutions*. Addison-Wesley.

Liquibase. (n.d.). *Liquibase documentation*. Retrieved October 9, 2026, from https://docs.liquibase.com/

Mermaid. (n.d.). *Mermaid: Diagramming and charting tool*. Retrieved October 9, 2026, from https://mermaid.js.org/

Microsoft. (n.d.-a). *.NET documentation*. Retrieved October 9, 2026, from https://learn.microsoft.com/dotnet/

Microsoft. (n.d.-b). *SQL Server technical documentation*. Retrieved October 9, 2026, from https://learn.microsoft.com/sql/sql-server/

Newman, S. (2021). *Building microservices: Designing fine-grained systems* (2nd ed.). O'Reilly Media.

RabbitMQ. (n.d.). *RabbitMQ documentation*. Broadcom. Retrieved October 9, 2026, from https://www.rabbitmq.com/docs

React Native. (n.d.). *React Native documentation*. Meta Platforms. Retrieved October 9, 2026, from https://reactnative.dev/docs/getting-started

Spring. (n.d.). *Spring Boot reference documentation*. Broadcom. Retrieved October 9, 2026, from https://docs.spring.io/spring-boot/